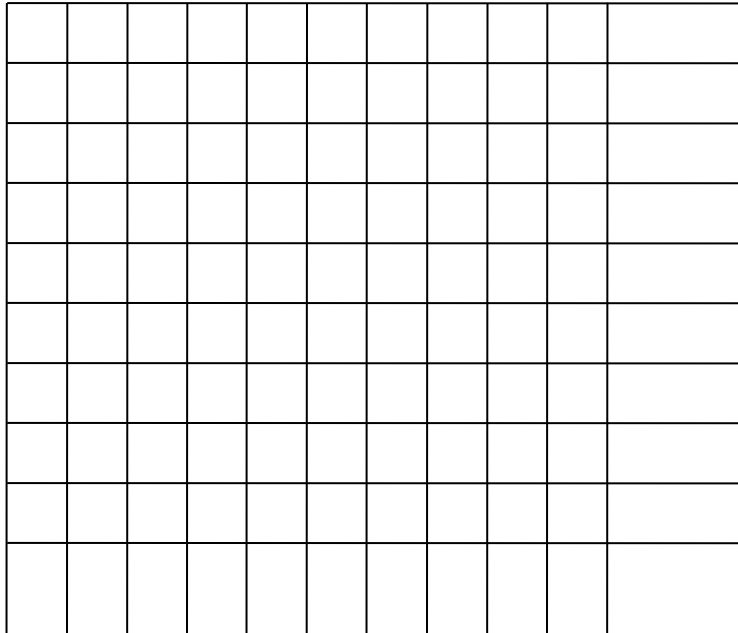


Počítačová grafika – PGR

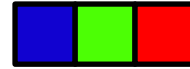
Opakování

prof. Ing. **Adam Herout**, PhD.





8bitů grayscale



24bitů RGB



32bitů RGBA



16bitů RGB

```
#define WIDTH 640
#define HEIGHT 480

static struct { char r,g,b; } image[WIDTH*HEIGHT];

#define Pixel(img, x, y)    img[WIDTH*y + x]

int main()
{
    ...
    Pixel(image, 30, 10).r = 150;
    ...
}
```

```
#define WIDTH 640
#define HEIGHT 480

static struct { char r,g,b; } image[WIDTH*HEIGHT];

#define Pixel(img, x, y)    img[WIDTH*y + x]

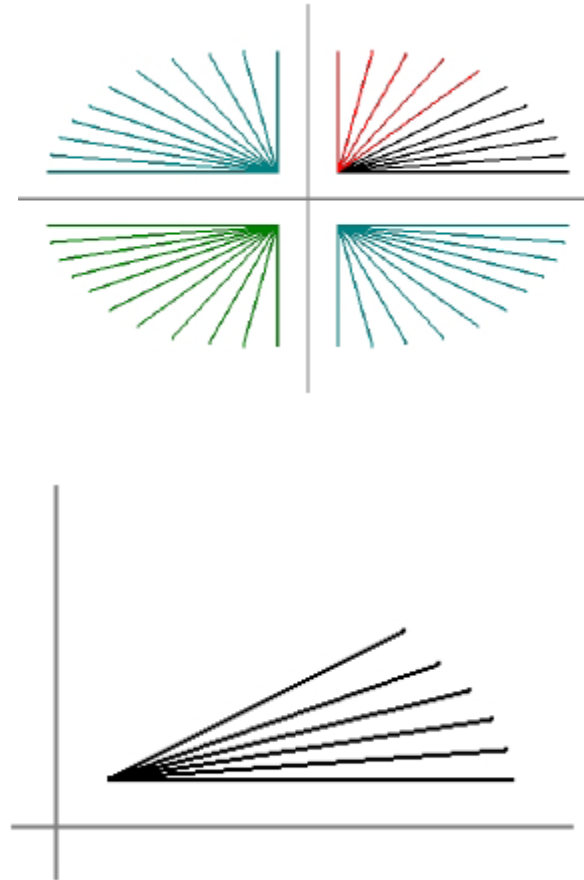
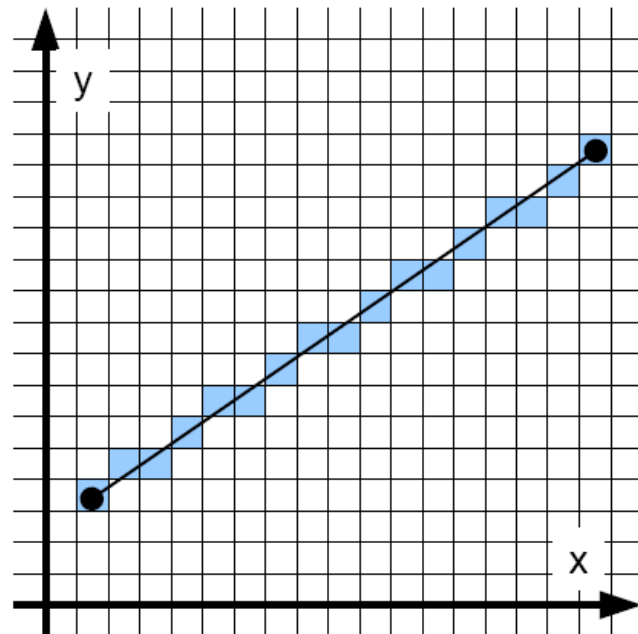
int main()
{
    memset(image, 0, sizeof(image)); // ale jen černá!!
    Pixel(image, 30, 10).r = 150;
    ...
}
```

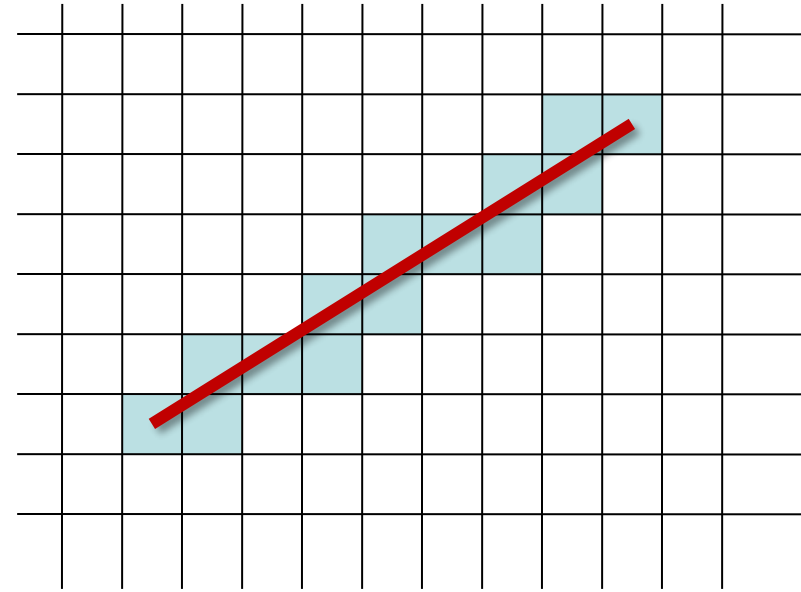
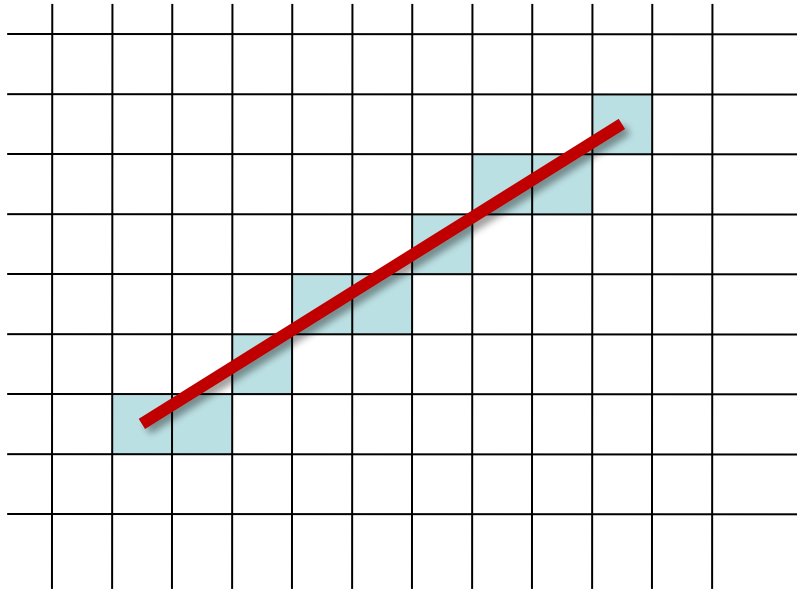
```
void FillRectangle(int x, int y, int w, int h, Color c)
{
    Color *p = &image[y*WIDTH + x];
    for (int i = 0; i < h; i++) {
        Color *pp = p;
        for (int j = 0; j < w; j++) {
            *pp++ = c;
        }
        p += WIDTH;
    }
}
```

```
// pozn: 320 = 140h = 101000000b
// 320x200 vs. 320x240
```

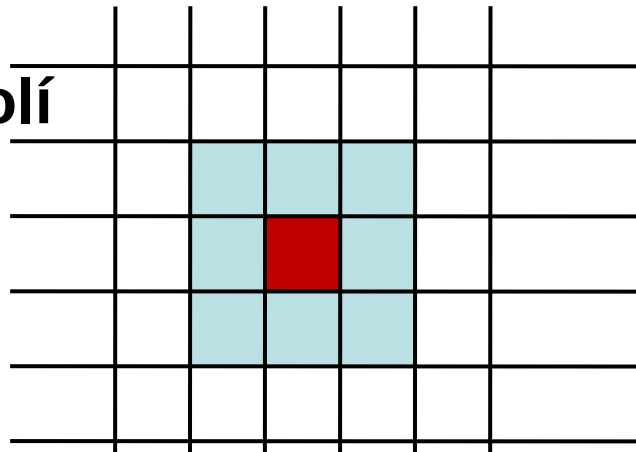
```
void DrawRectangle(int x, int y, int w, int h, Color c)
{
    Color *p = &image[y*WIDTH + x];
    Color *pp = &image[(y+h)*WIDTH + x];
    for (int i = 0; i < w; i++) {
        *p++ = c;    *pp++ = c;
    }
    p = &image[y*WIDTH + x];
    pp = &image[y*WIDTH + x + w];
    for (int i = 0; i < h; i++) {
        *p = c;      *pp = c;
        p += WIDTH;  pp += WIDTH;
    }
}
```

- Úsečka daná:
 - Dvěma body (x,y)
- Možnost přeskládat do prvního oktantu

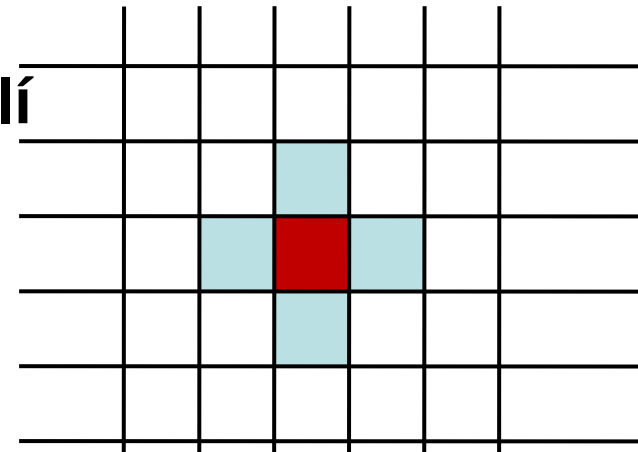




8-sousednost



4-sousednost




```
void DrawLine(int x1, int y1, int x2, int y2, Color c)
{
    float y = y1;
    float dy = (y2-y1) / (x2-x1);
    for (int x = x1; x <= x2; x++) {
        image[int(y)*WIDTH + x] = c;
        y += dy;
    }
}
```

```
void DrawLine(int x1, int y1, int x2, int y2, Color c)
{
    float E = y1 - 0.5;
    float dy = (y2-y1) / (x2-x1);
    Color *p = &image[y*WIDTH + x];
    for (int x = x1; x <= x2; x++) {
        *p++ = c; // draw_pixel(...)
        E += dy;
        if (E >= 0.0) { p += WIDTH; E -= 1.0; }
    }
}
```

```
void DrawLine(int x1, int y1, int x2, int y2, Color c)
{
    int E = (y1 - 0.5) * (x2-x1);
    int dy = (y2-y1); // ubylo / (x2-x1)
    Color *p = &image[y1*WIDTH + x1];
    for (int x = x1; x <= x2; x++) {
        *p++ = c; // draw_pixel(...)
        E += dy;
        if (E >= 0.0) { p += WIDTH; E -= (x2-x1); }
    }
}
```

Floating point:



$$x = S \cdot M \cdot 2^E$$

abcdefghijklmnpq

=0,abcdefghijklmnpq

$$=a \cdot 1/2 + b \cdot 1/4 + c \cdot 1/8 + d \cdot 1/16 + \dots$$

Fixed point:

qponmlkjihgfedcba

=qponmlkjihgfedcba,0

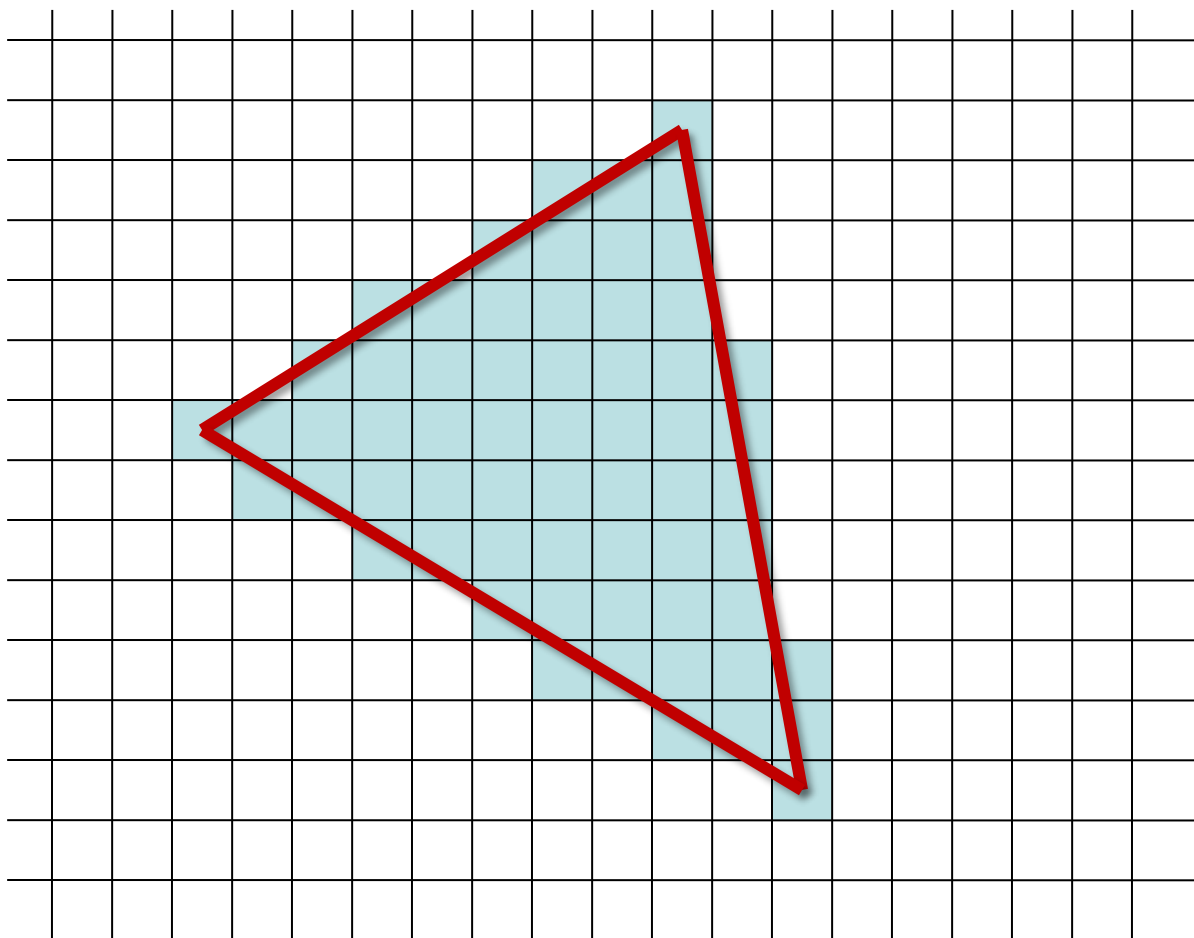
$$=a \cdot 1 + b \cdot 2 + c \cdot 4 + d \cdot 8 + \dots$$

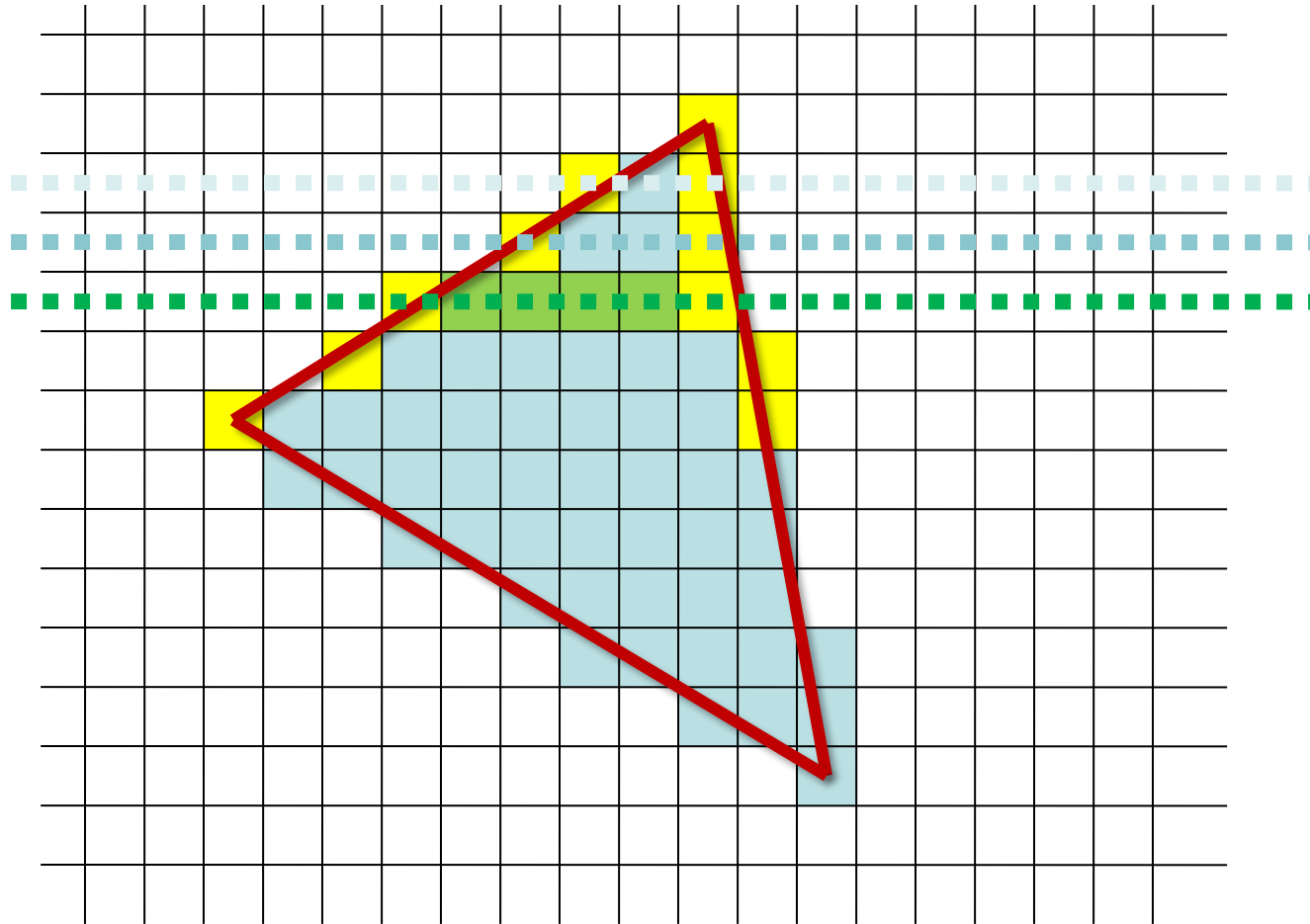
dcbaefghijklmnpq

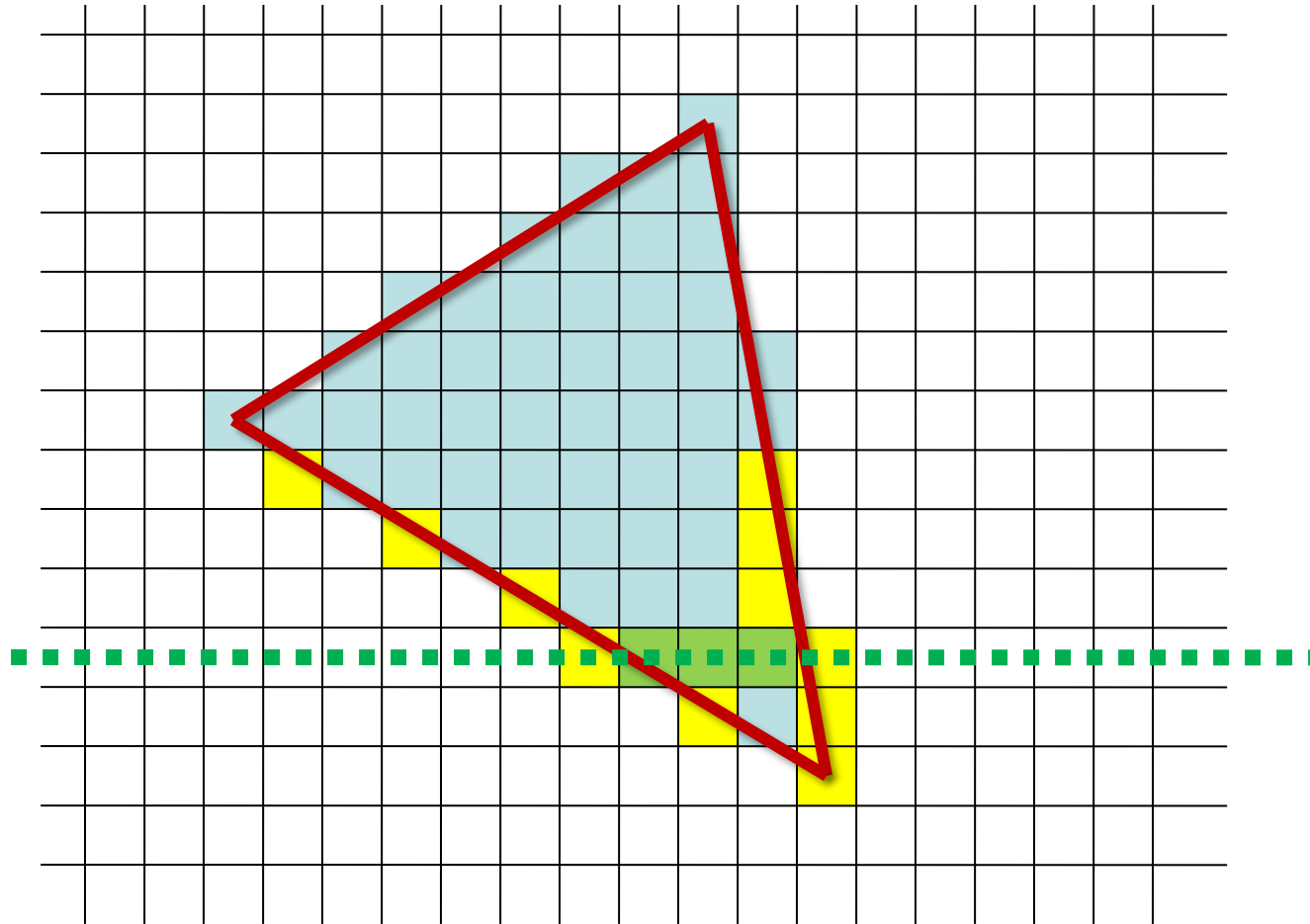
=dcba,efghijklmnpq

$$=a \cdot 1 + b \cdot 2 + c \cdot 3 + d \cdot 4 + e \cdot 1/2 + f \cdot 1/4 + g \cdot 1/8 + h \cdot 1/16 + \dots$$

```
#define FRAC_BITS 8
void DrawLine(int x1, int y1, int x2, int y2, Color c)
{
    int y = y1 << FRAC_BITS;
    int dy = (y2-y1) << FRAC_BITS / (x2-x1);
    for (int x = x1; x <= x2; x++) {
        image[(y >> FRAC_BITS)*WIDTH + x] = c;
        y += dy;
    }
}
```

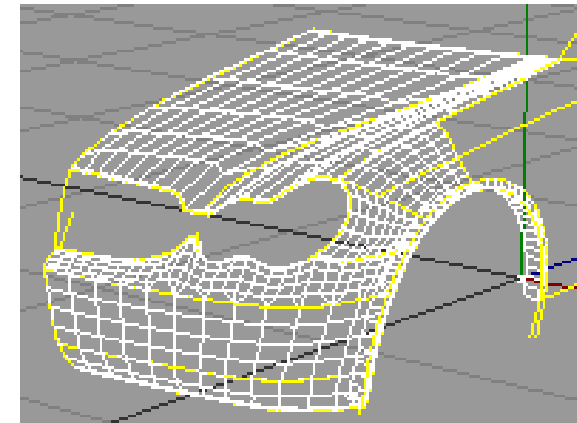
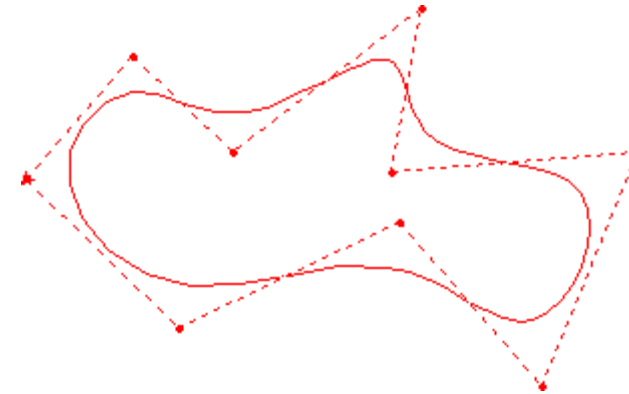


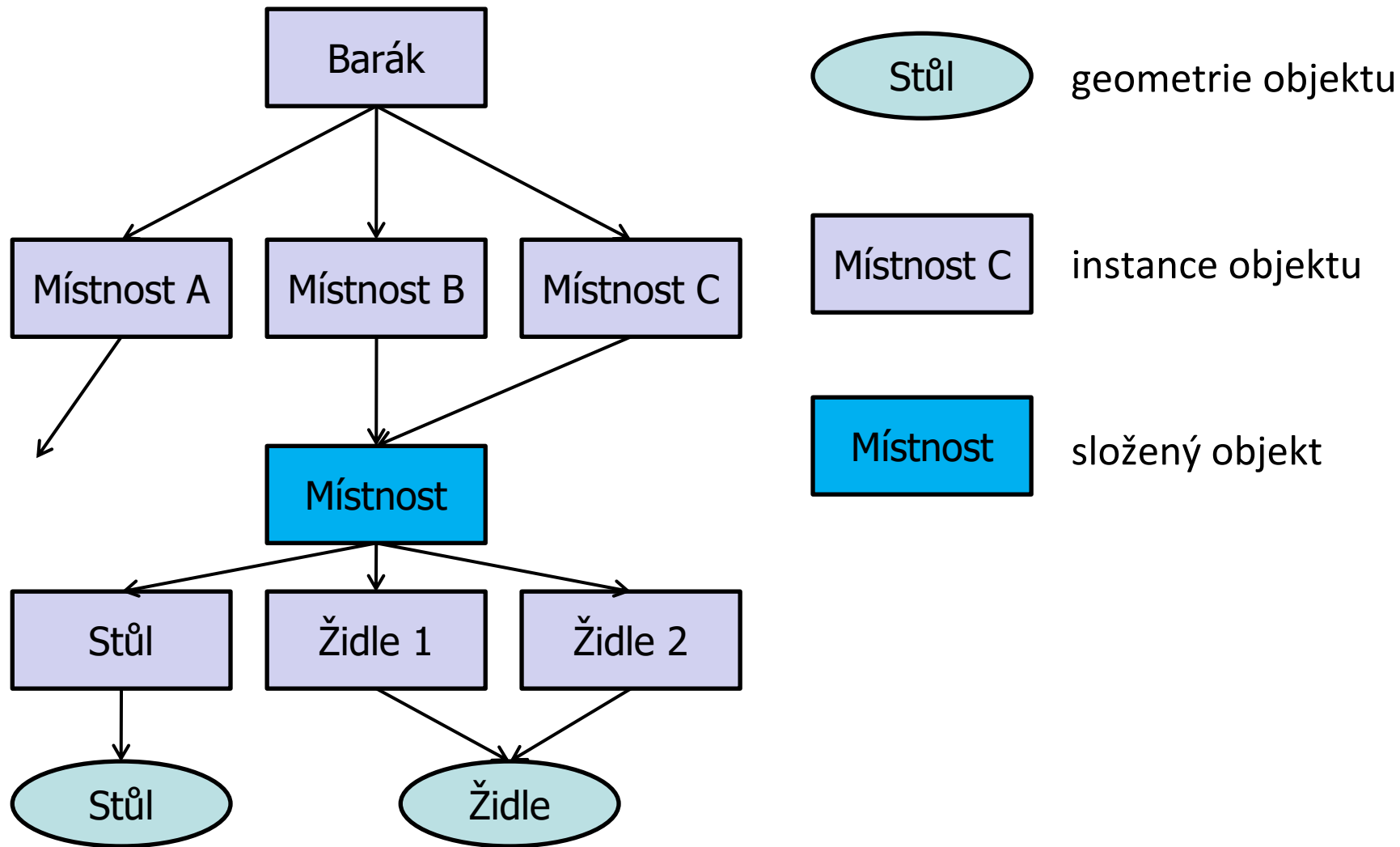




1. Určit nejvyšší vrchol
2. Vypočíst parametry DDA pro obě hrany,
Určit výšku nižší hrany
3. Vykreslit příslušný počet řádků (do dalšího vrcholu)
4. Vypočíst parametry DDA pro zbývající hranu
5. Vykreslit řádky až do konce

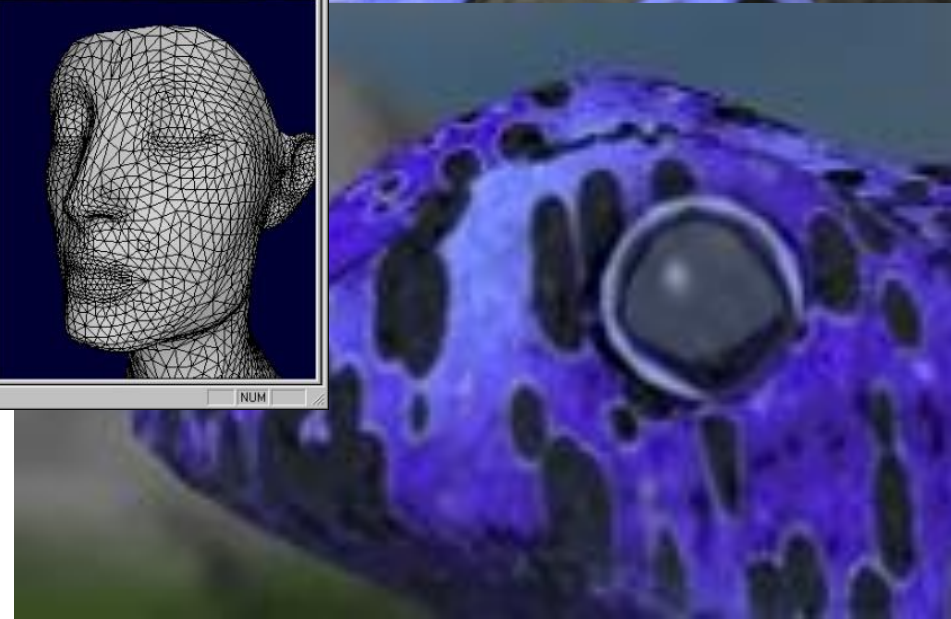
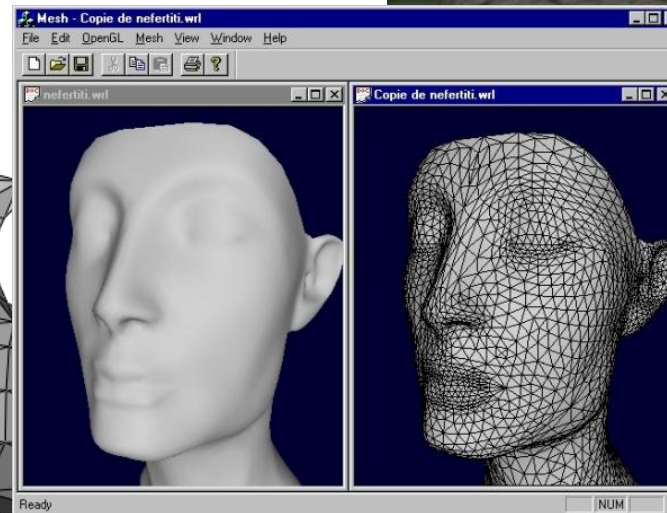
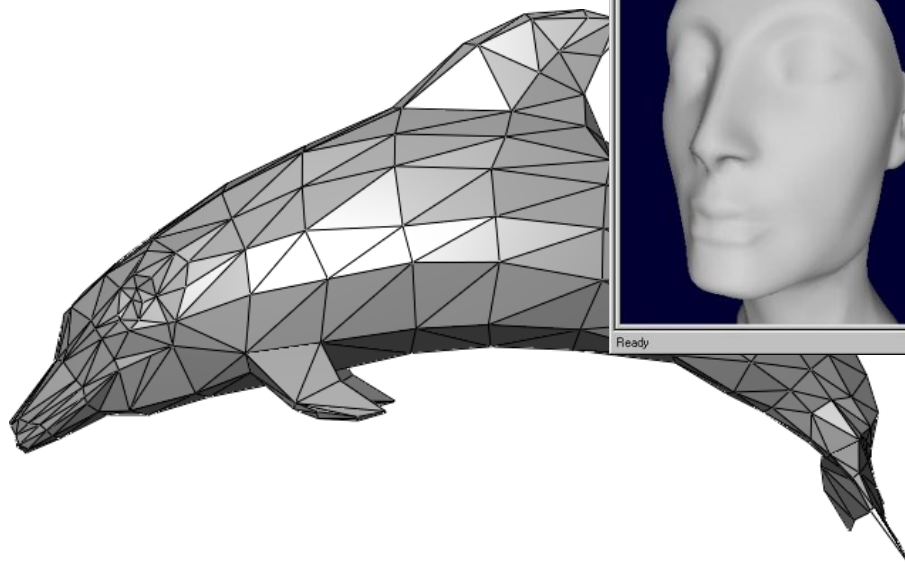
- Křivky
 - Řídící body
 - Vektor vah
 - Parametrické vyjádření:
 $(x,y,z) = f(t), 0.0 \leq t \leq 1.0$
 - Vykreslení: množství úseček
- Plochy
 - Podobně
 - Parametrické vyjádření:
 $(x,y,z) = f(u,v), 0.0 \leq u \leq 1.0, 0.0 \leq v \leq 1.0$
 - Vykreslení: množství čtyřúhelníků (??!!)

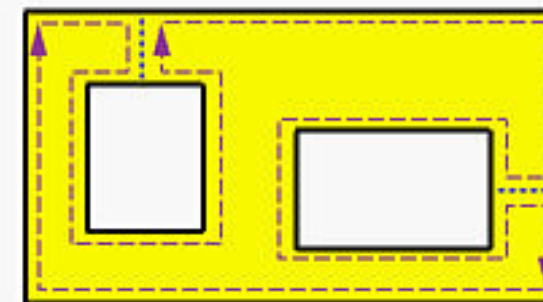
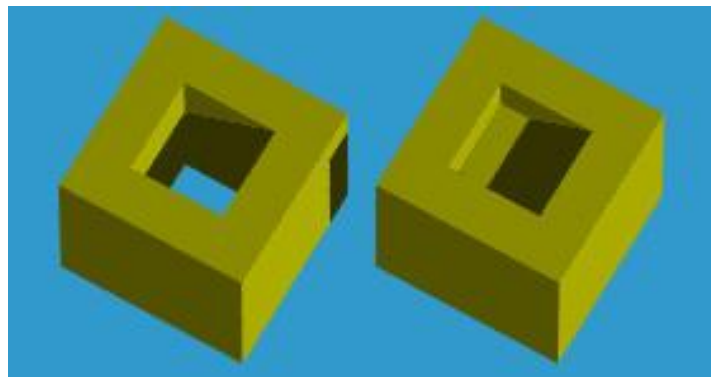
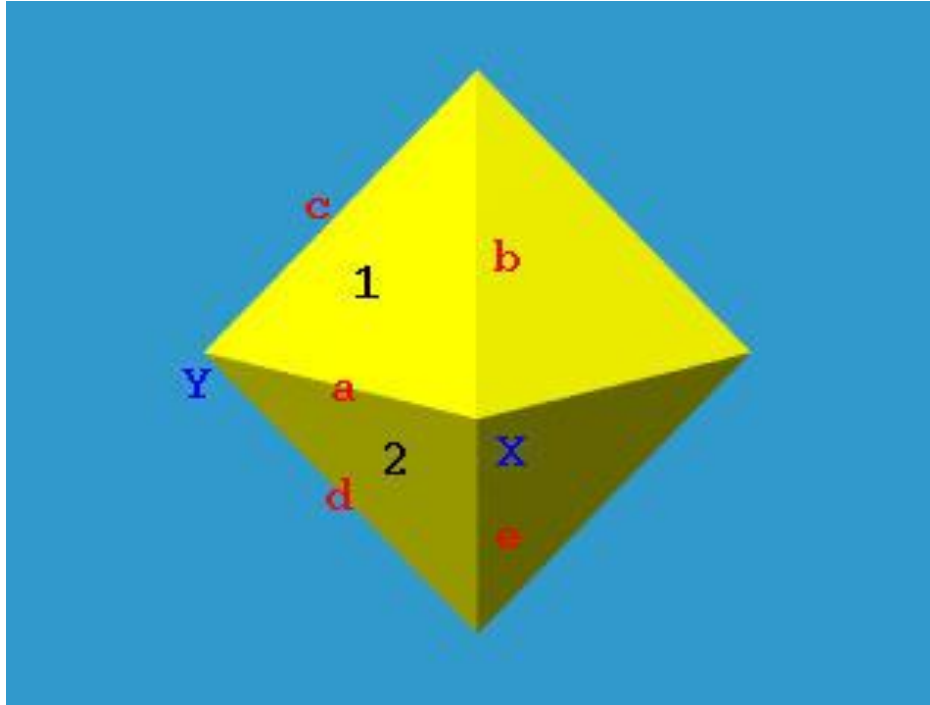
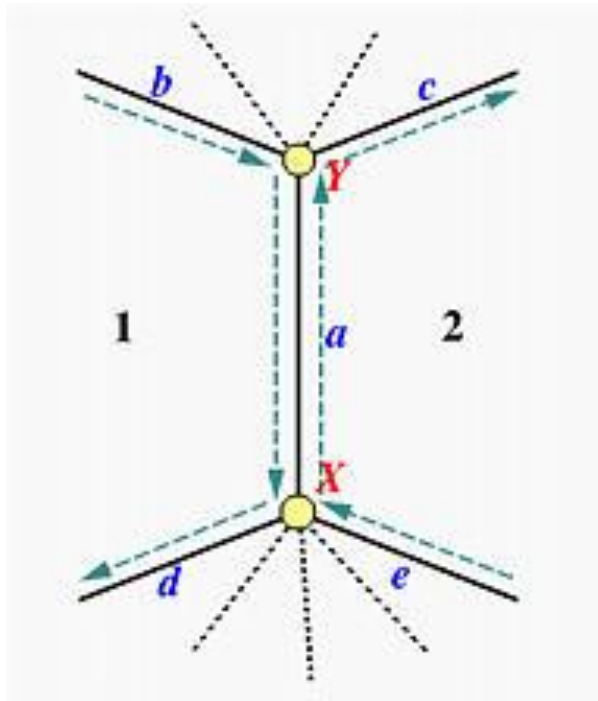




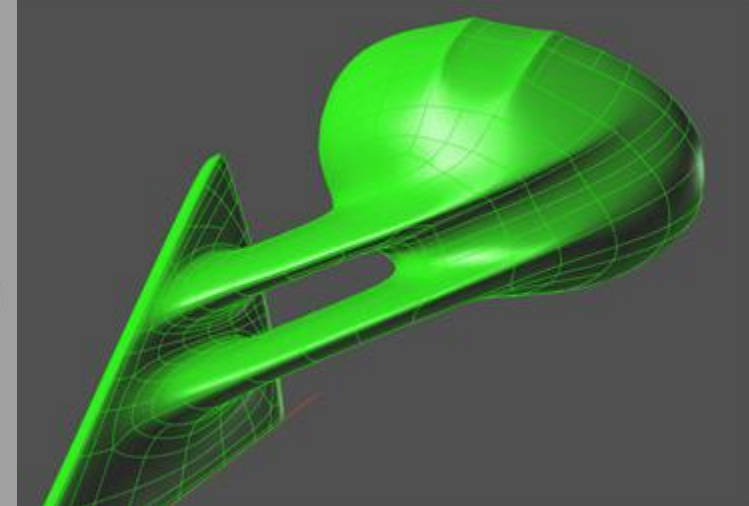
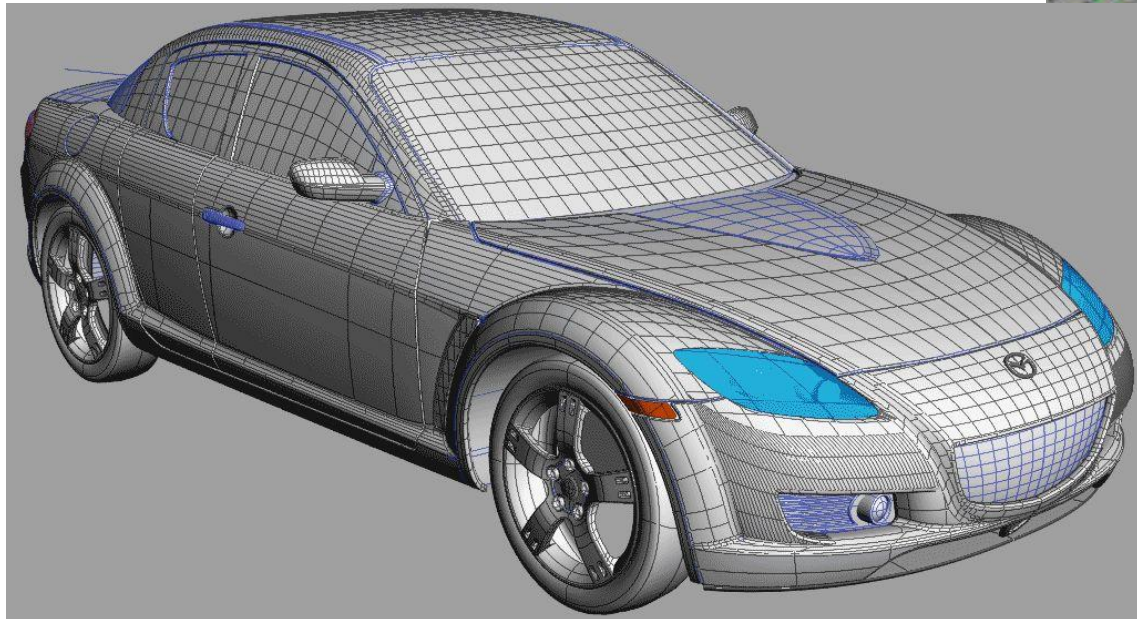
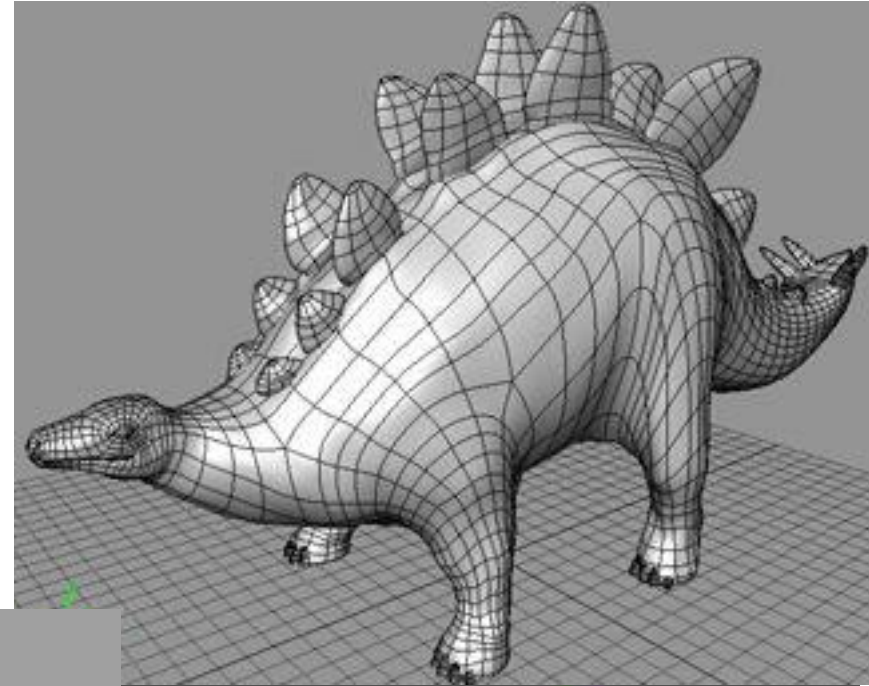
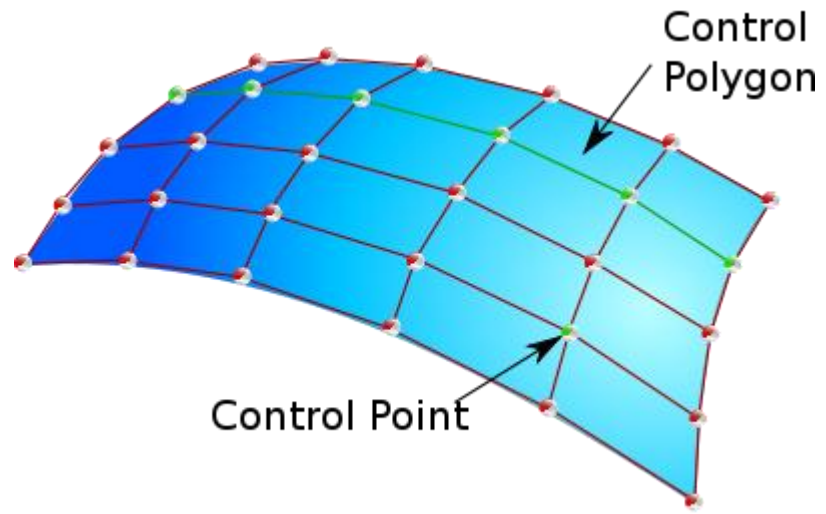


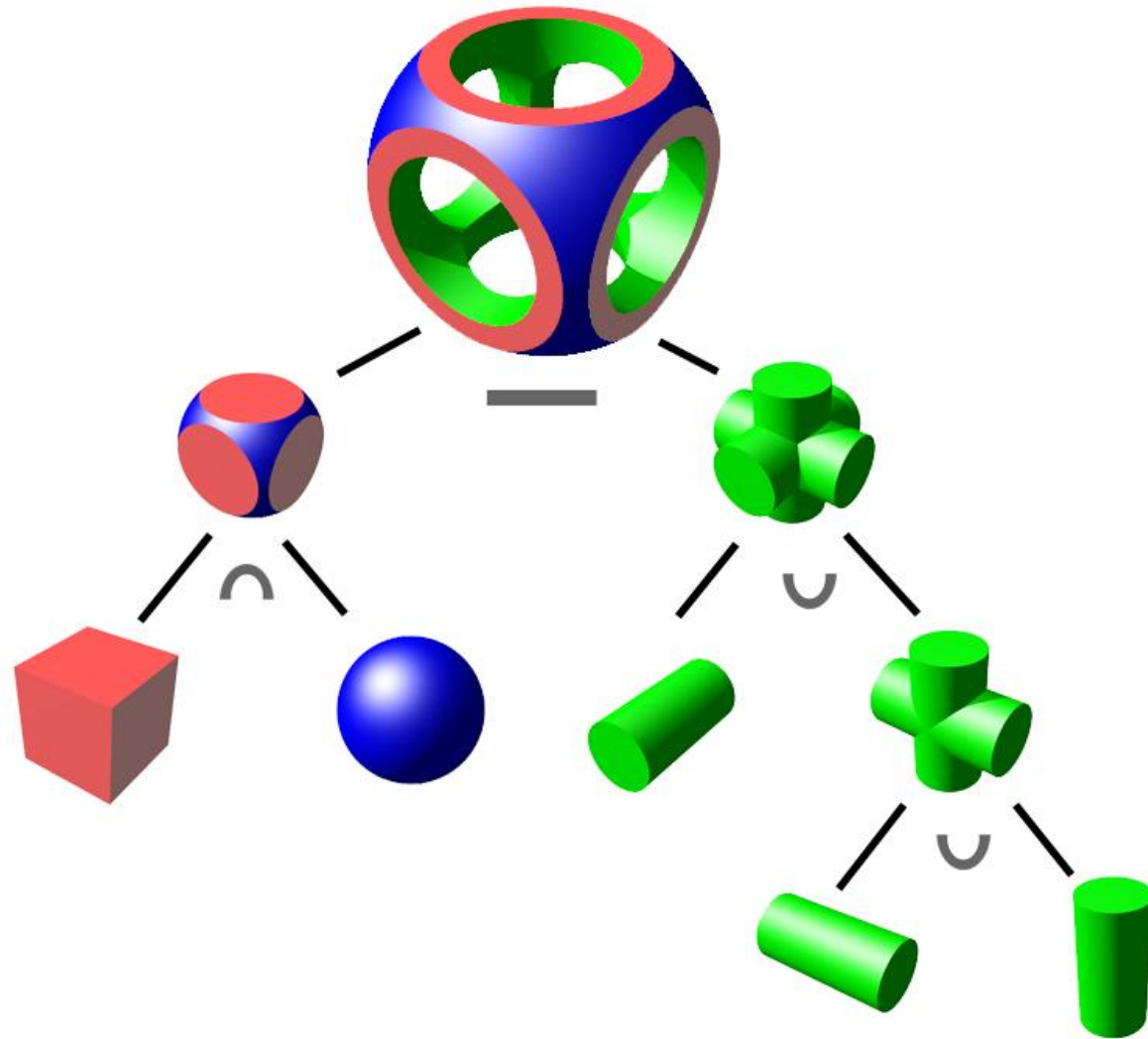
- strukturovaná vs. nestrukturovaná
- co trojúhelník to 3x3 souř.
- seznam vrcholů, seznam trojúhelníků

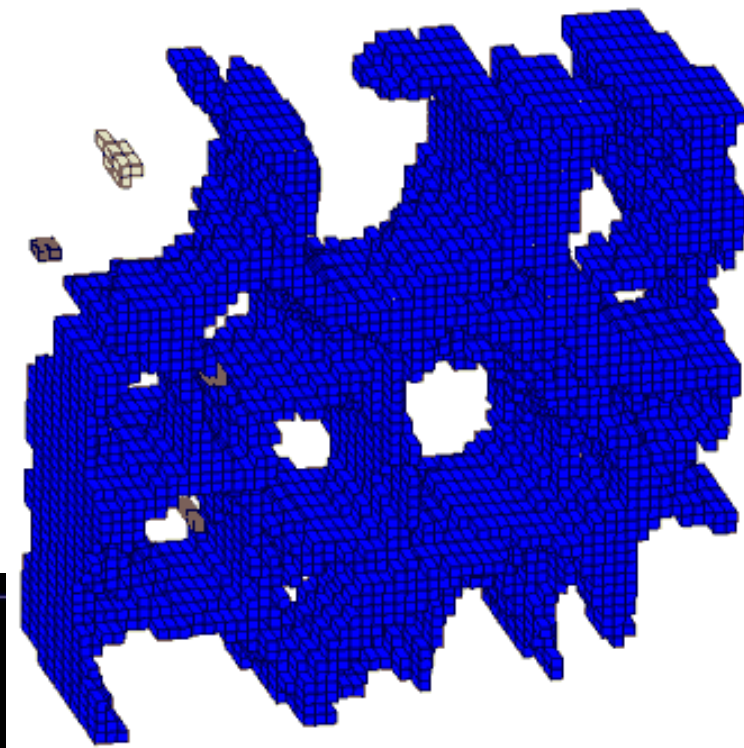
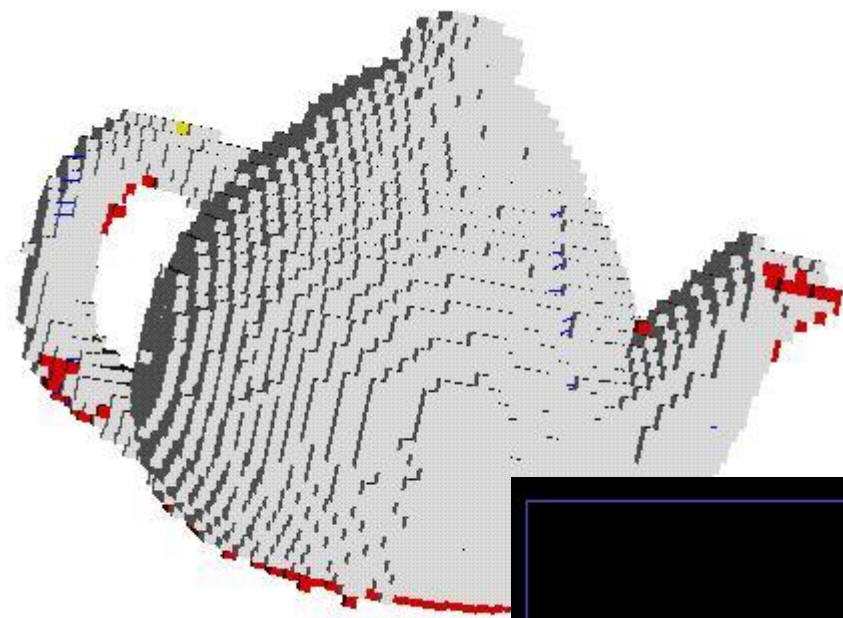


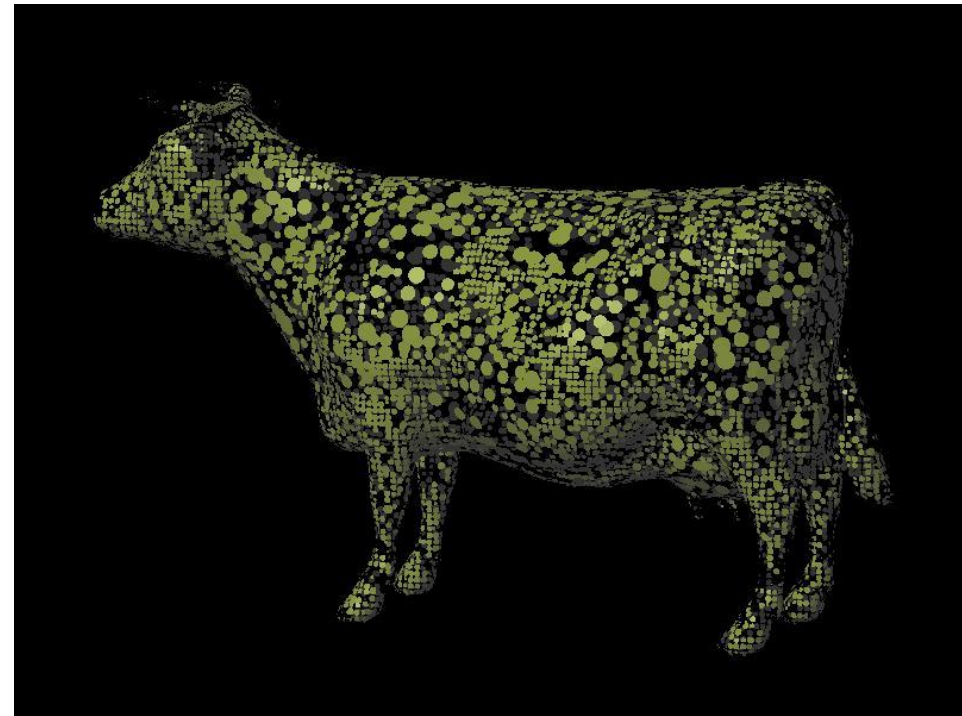


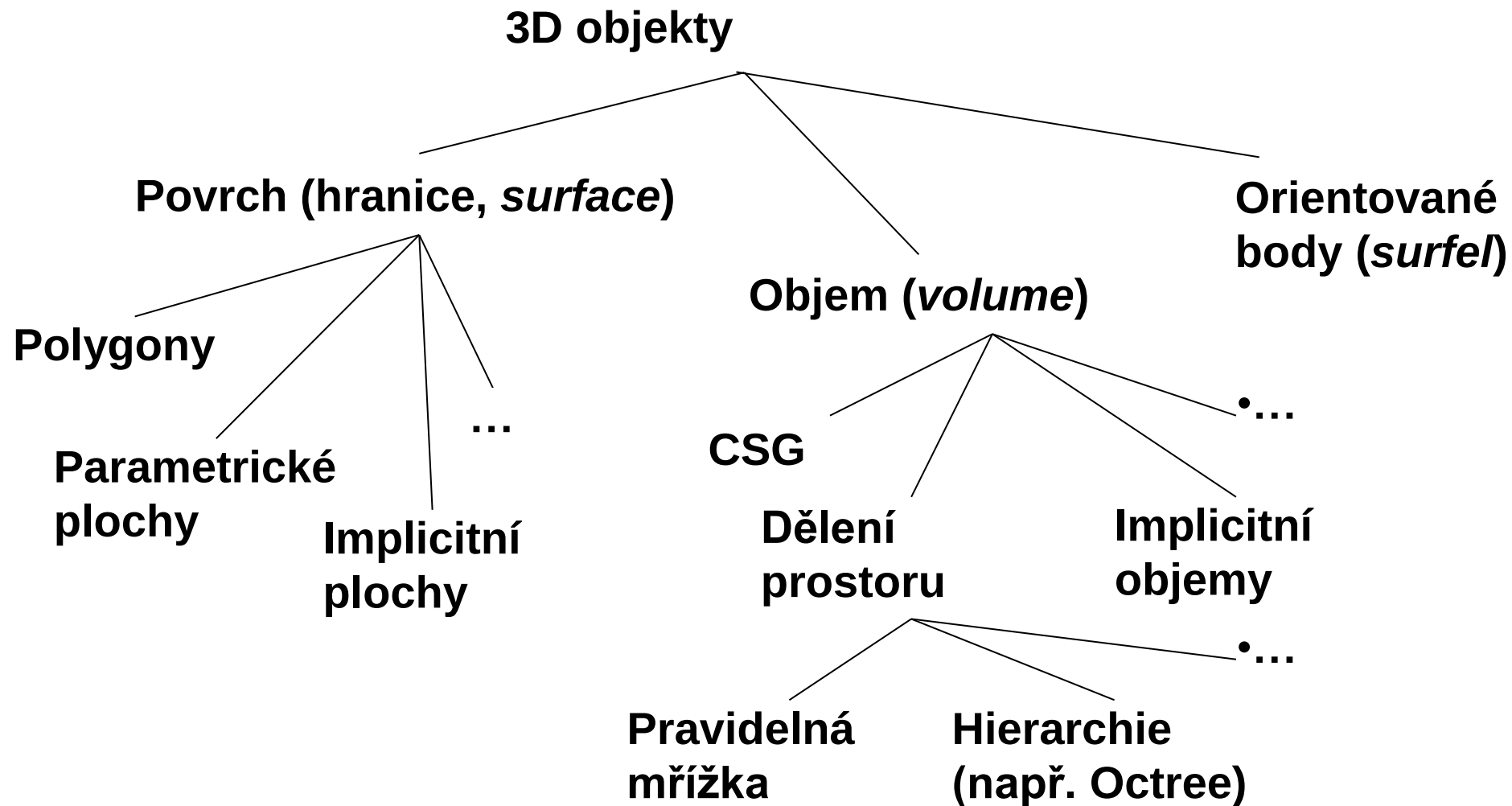
.....auxiliary edges







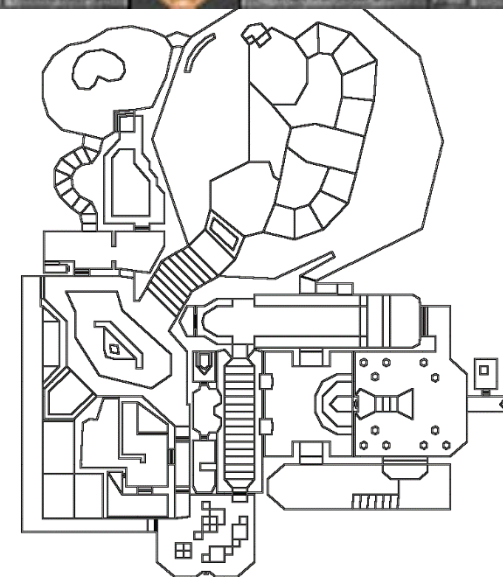
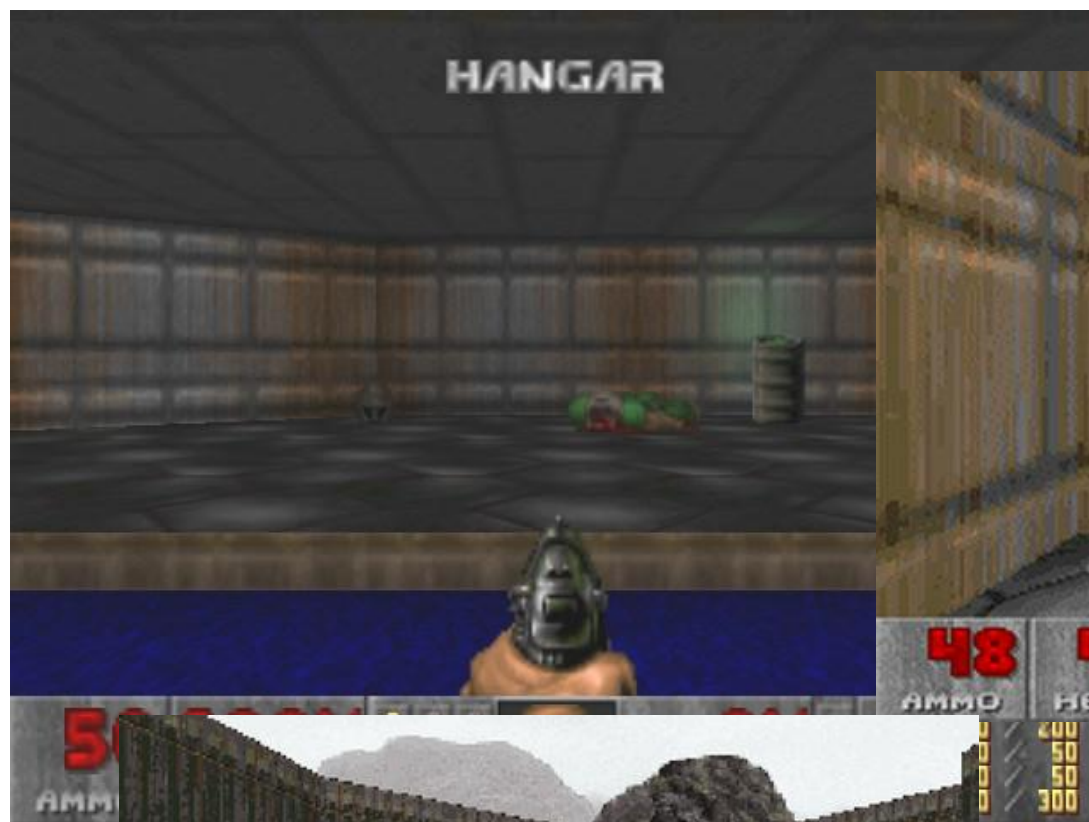




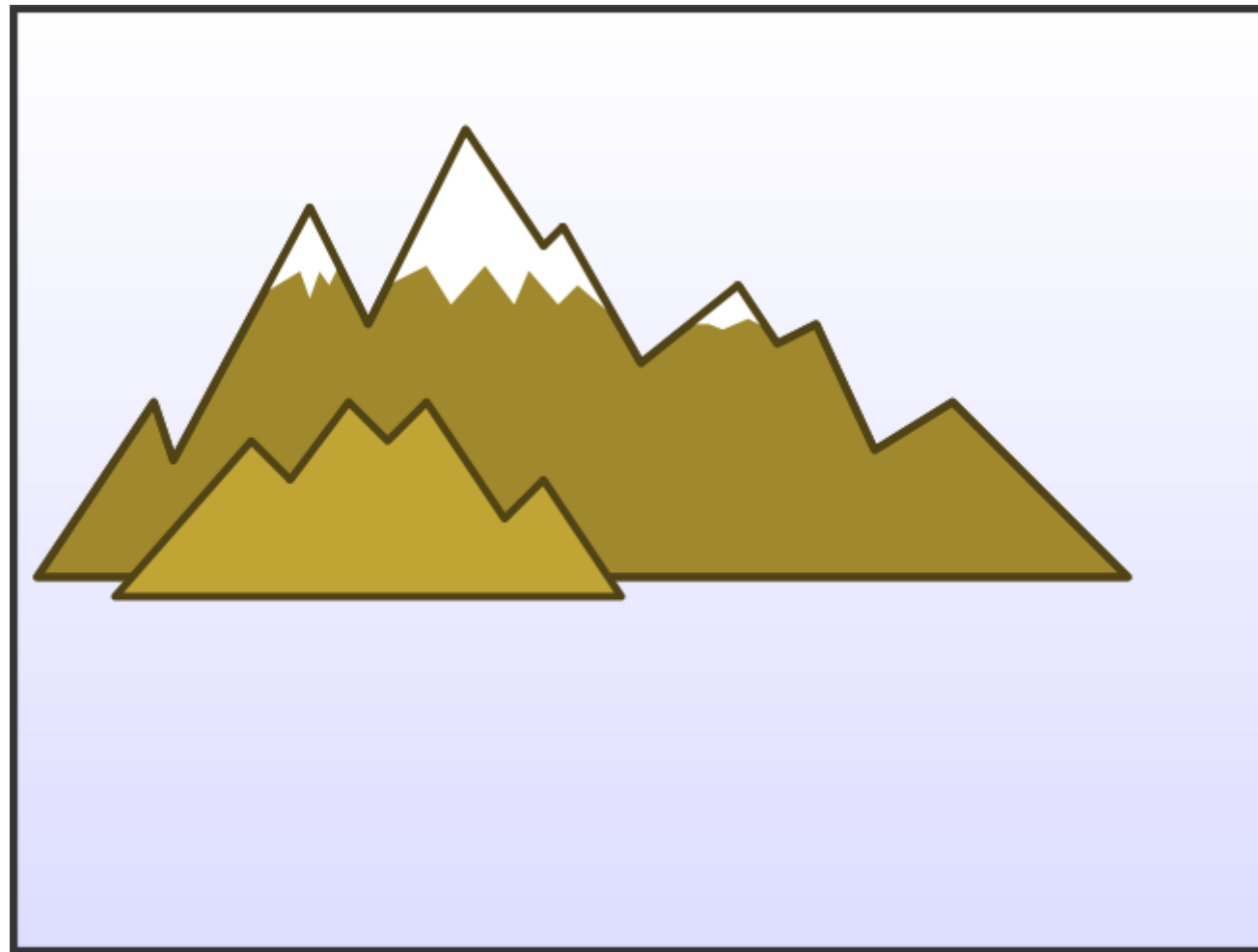


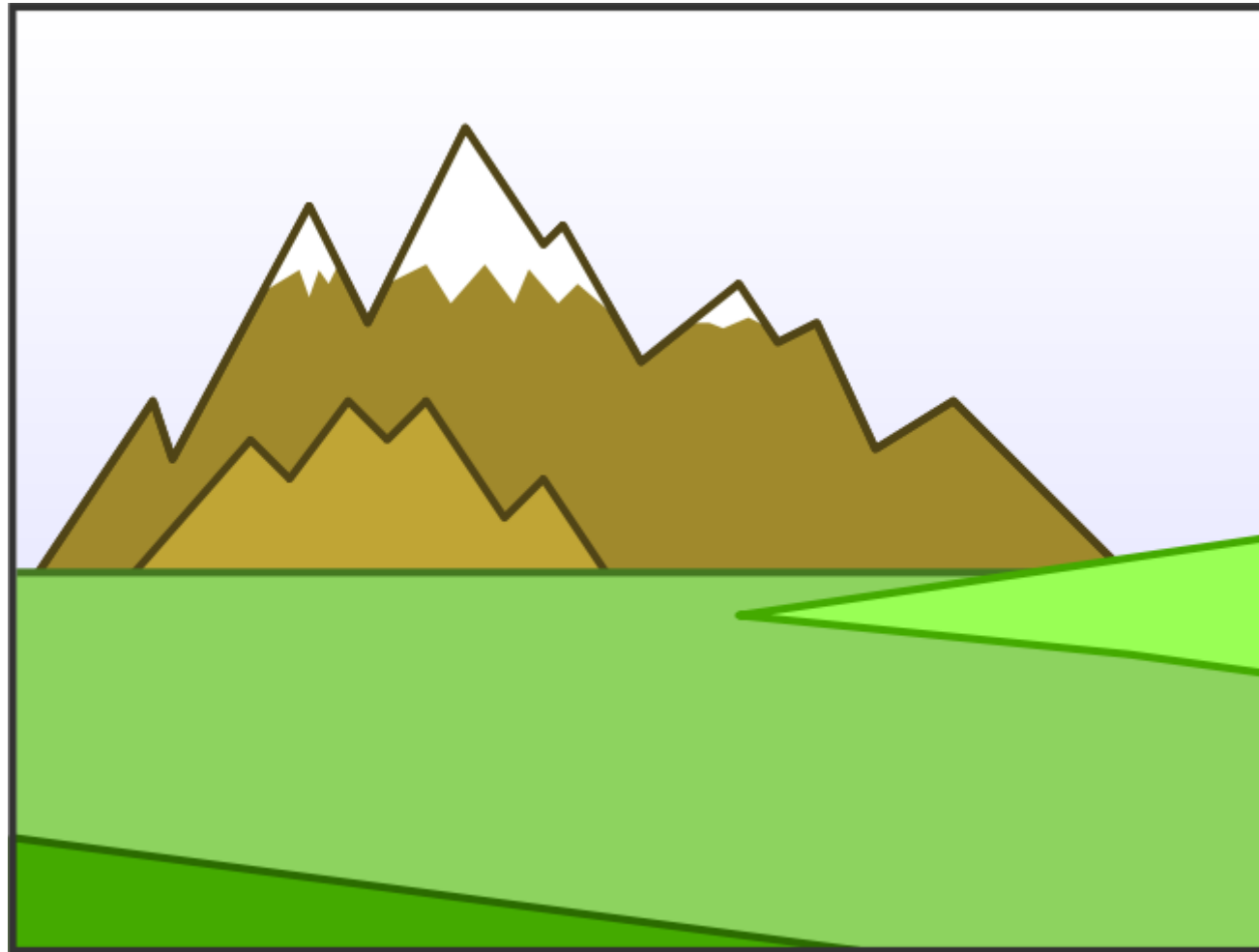


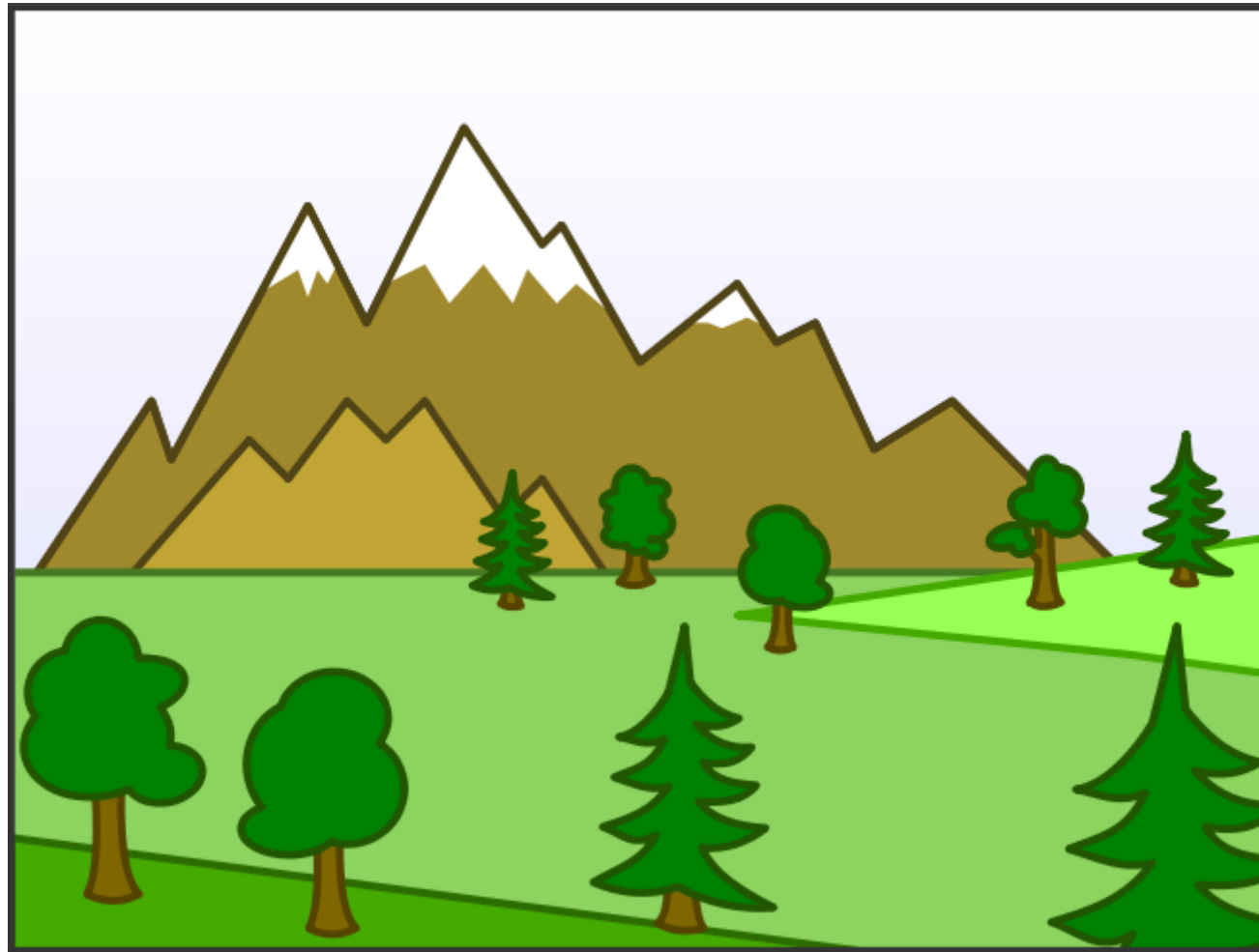
I 2D nebo 3D?







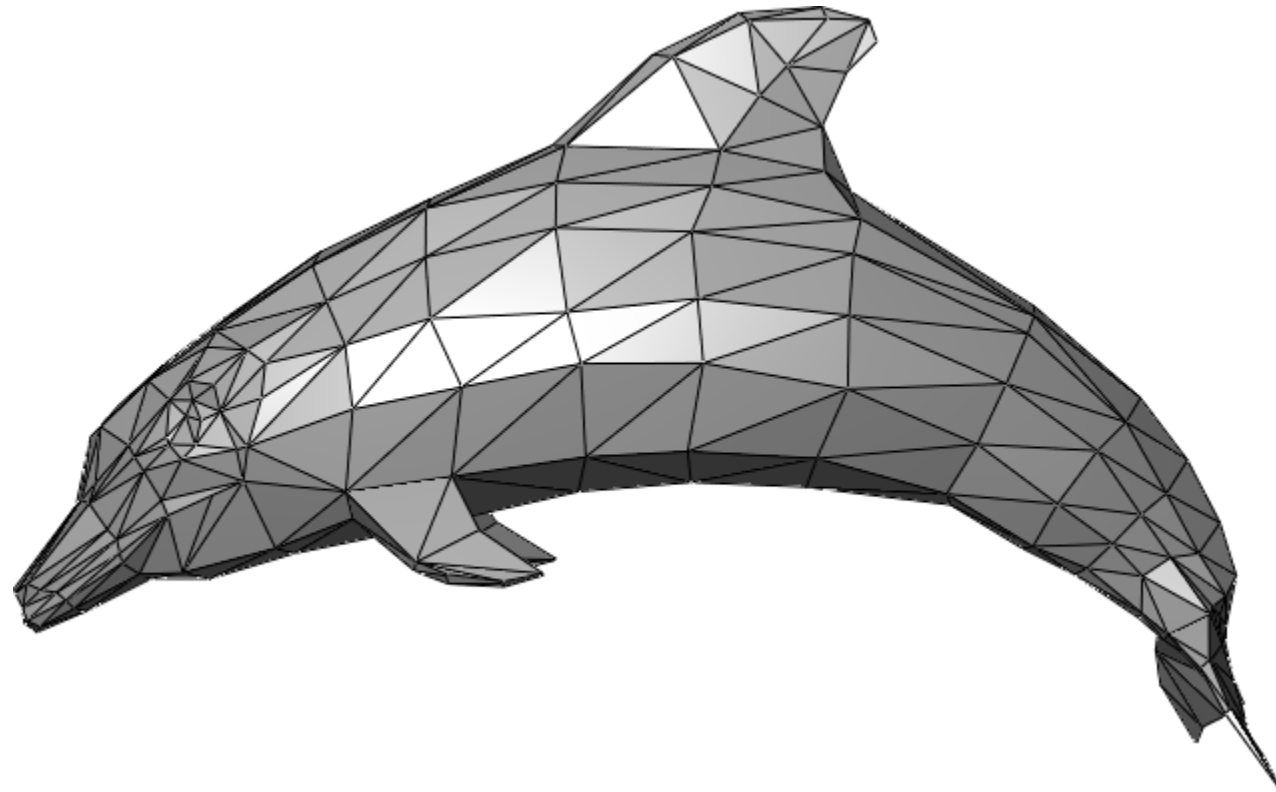




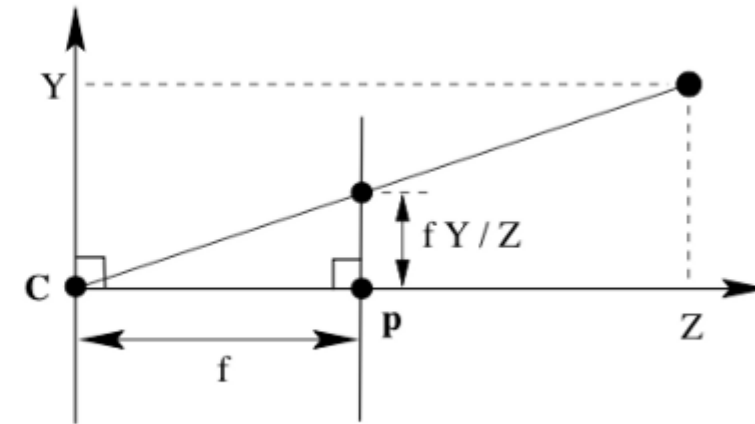
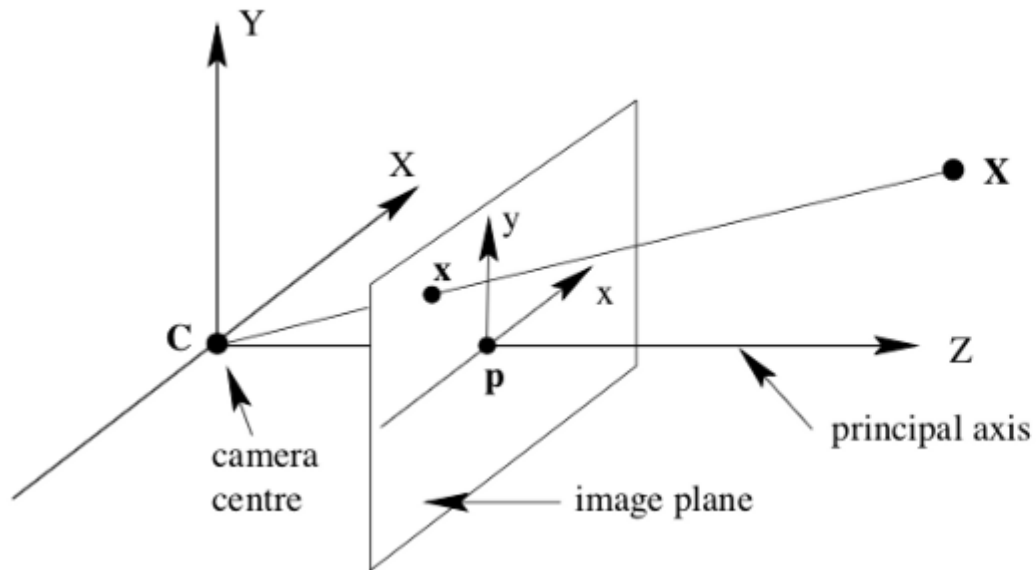
2D nebo 3D?



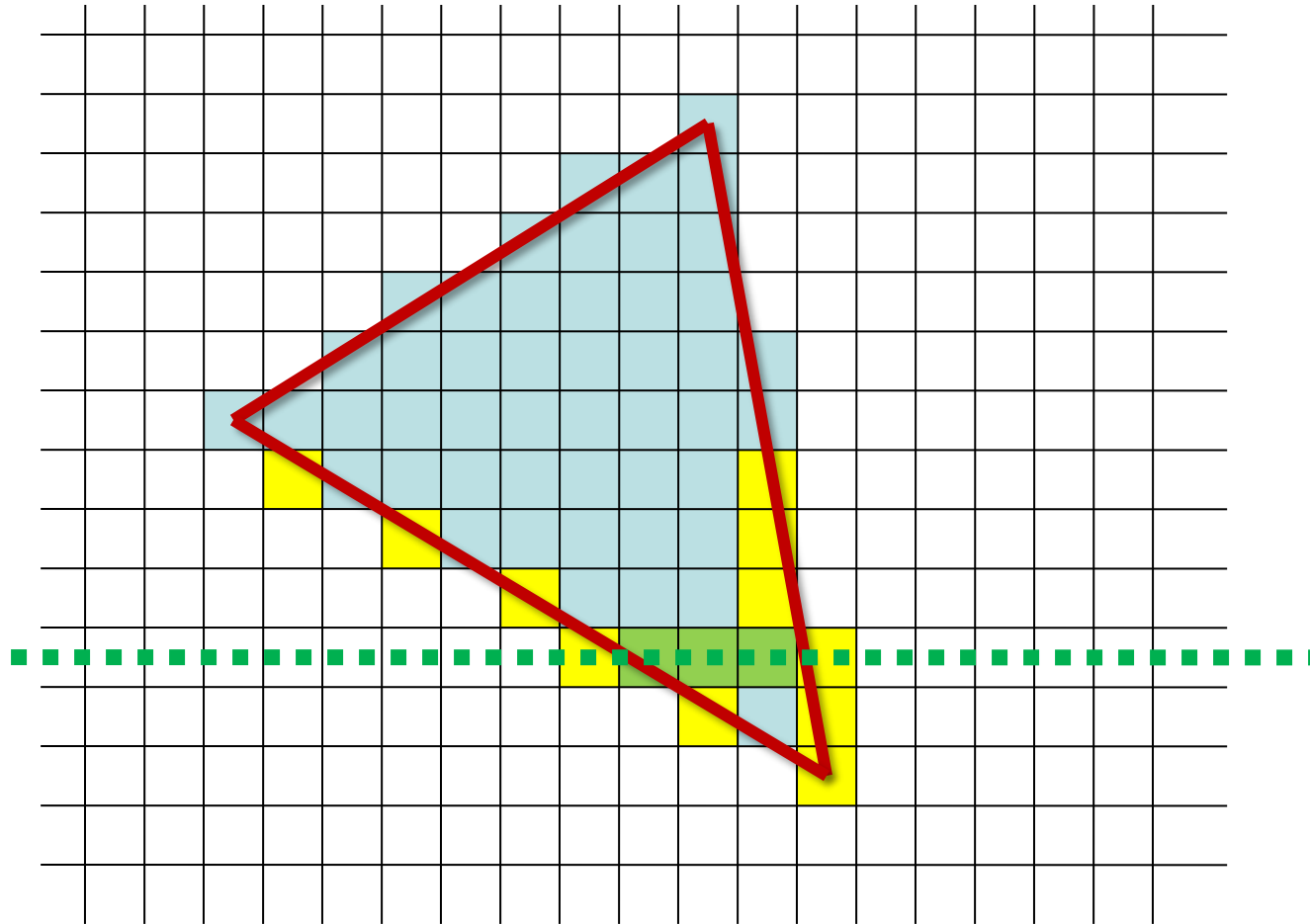




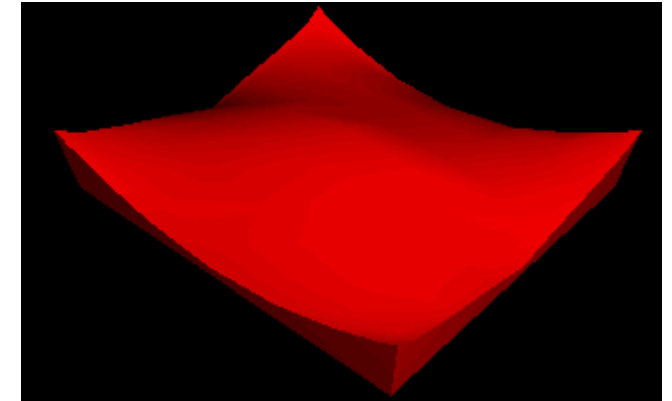
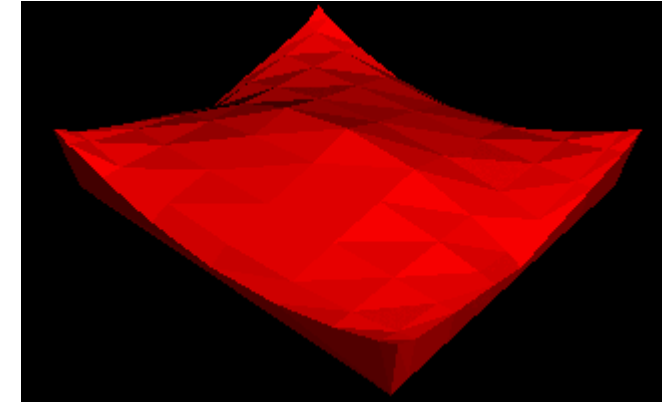
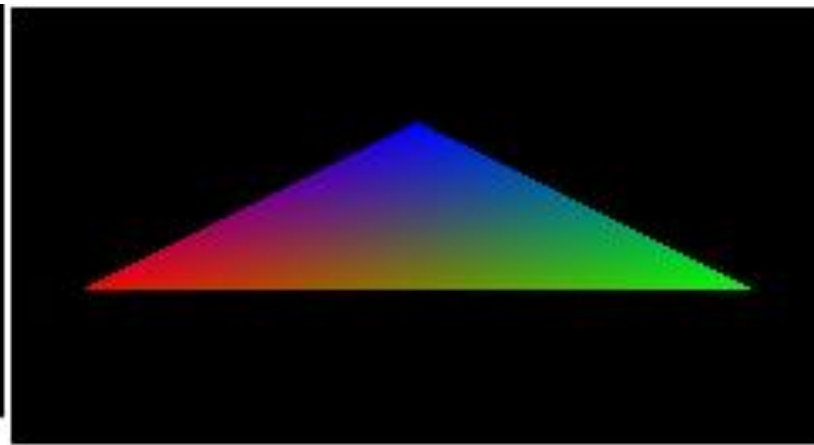
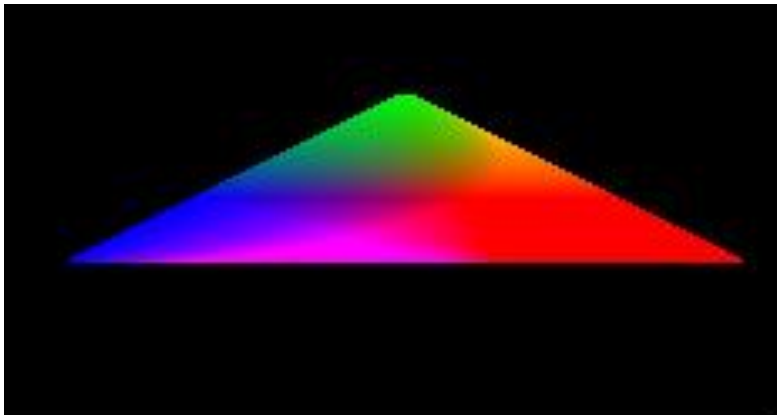
- Trojúhelník popsán: ...

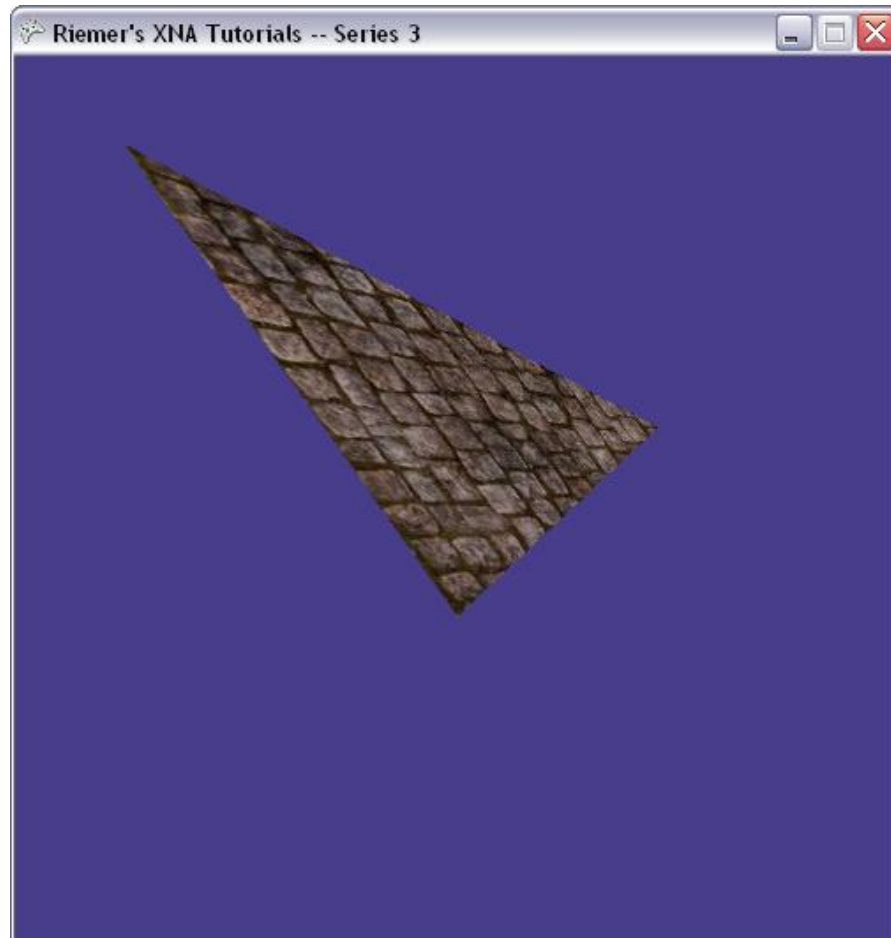


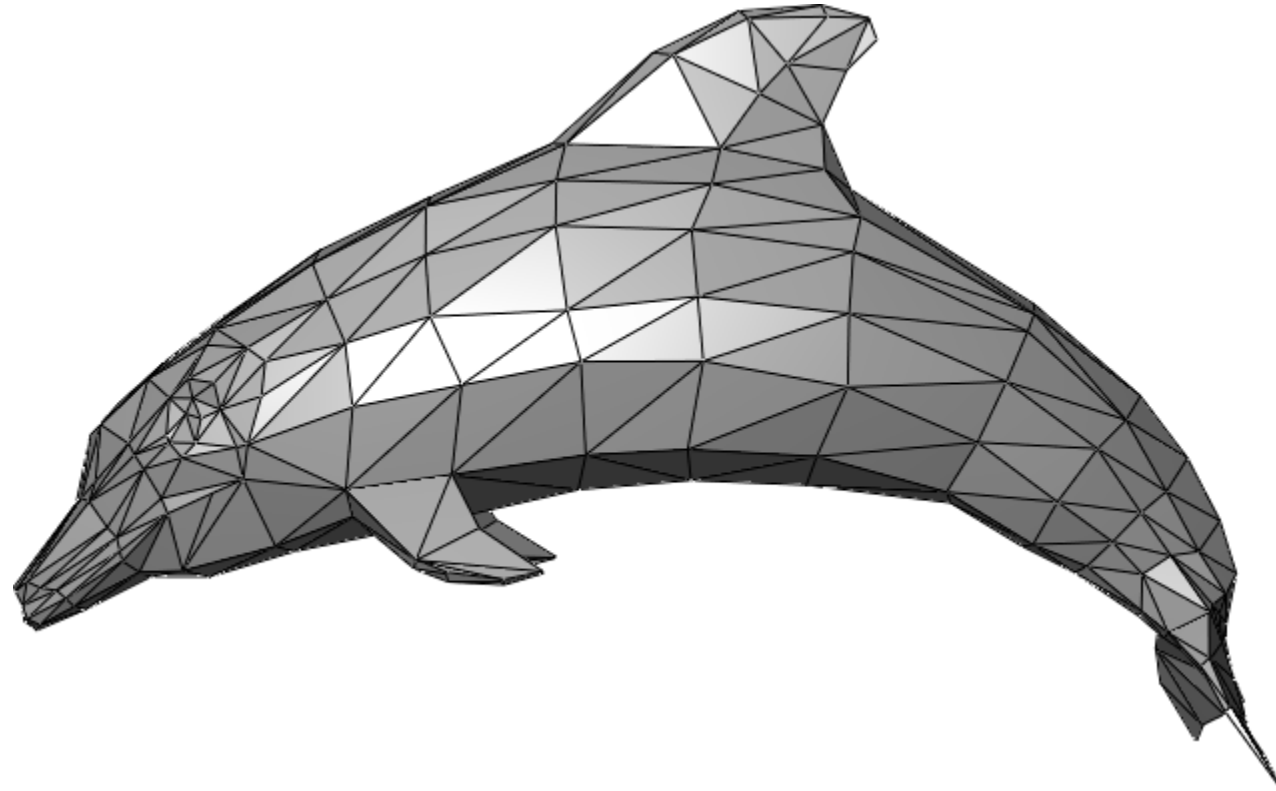
$$x' = Mx \quad \begin{bmatrix} x' \\ y' \\ 0 \\ w' \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



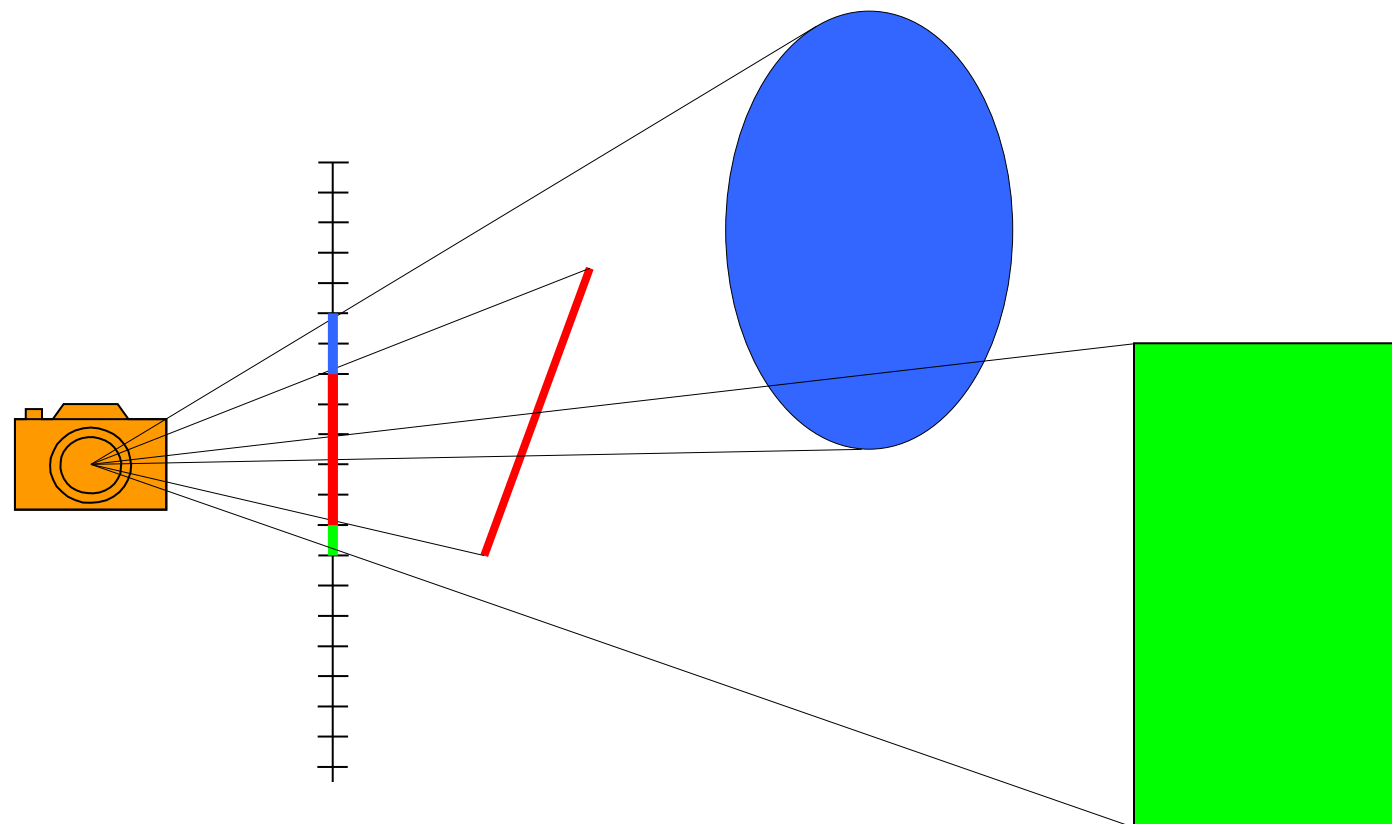
- Barvy
- Texturovacích souřadnic
- Hodnot osvětlení

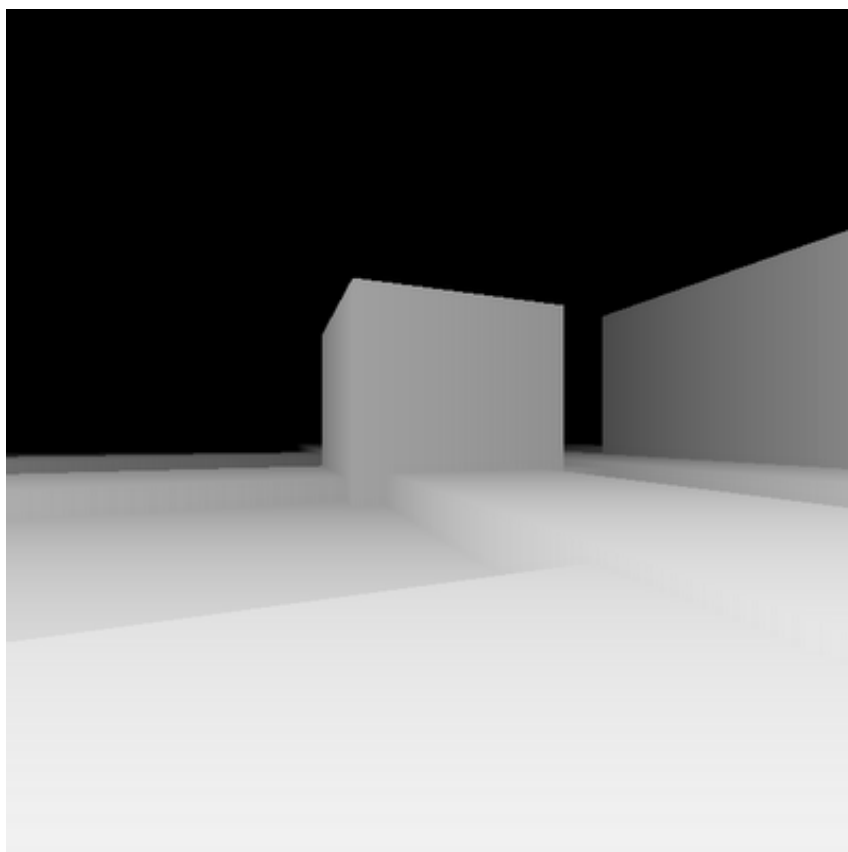






- Jenže: **viditelnost?**





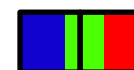
8bitů grayscale



24bitů RGB



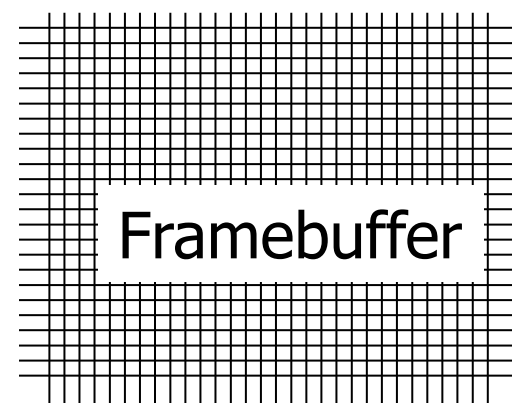
32bitů RGBA

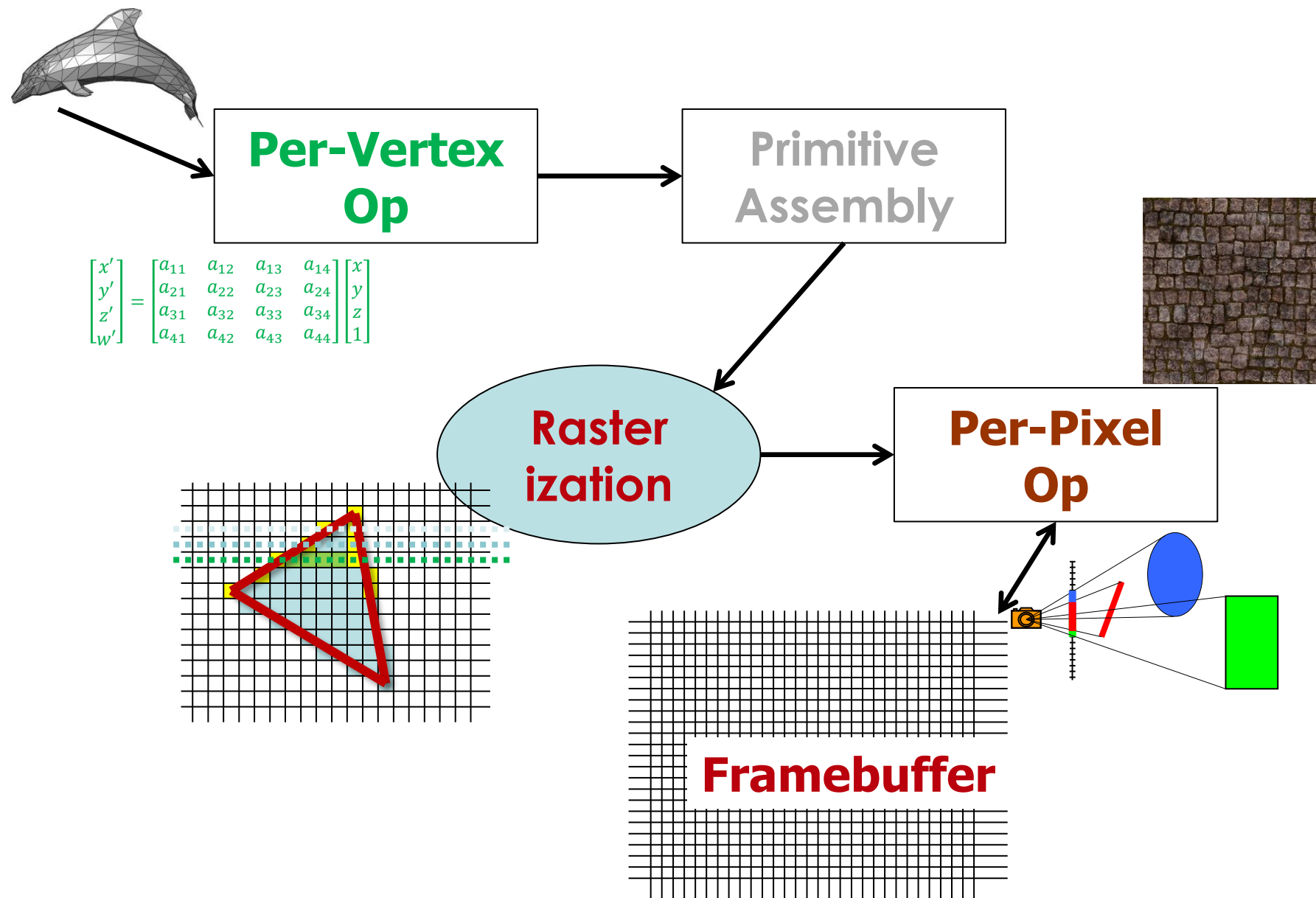


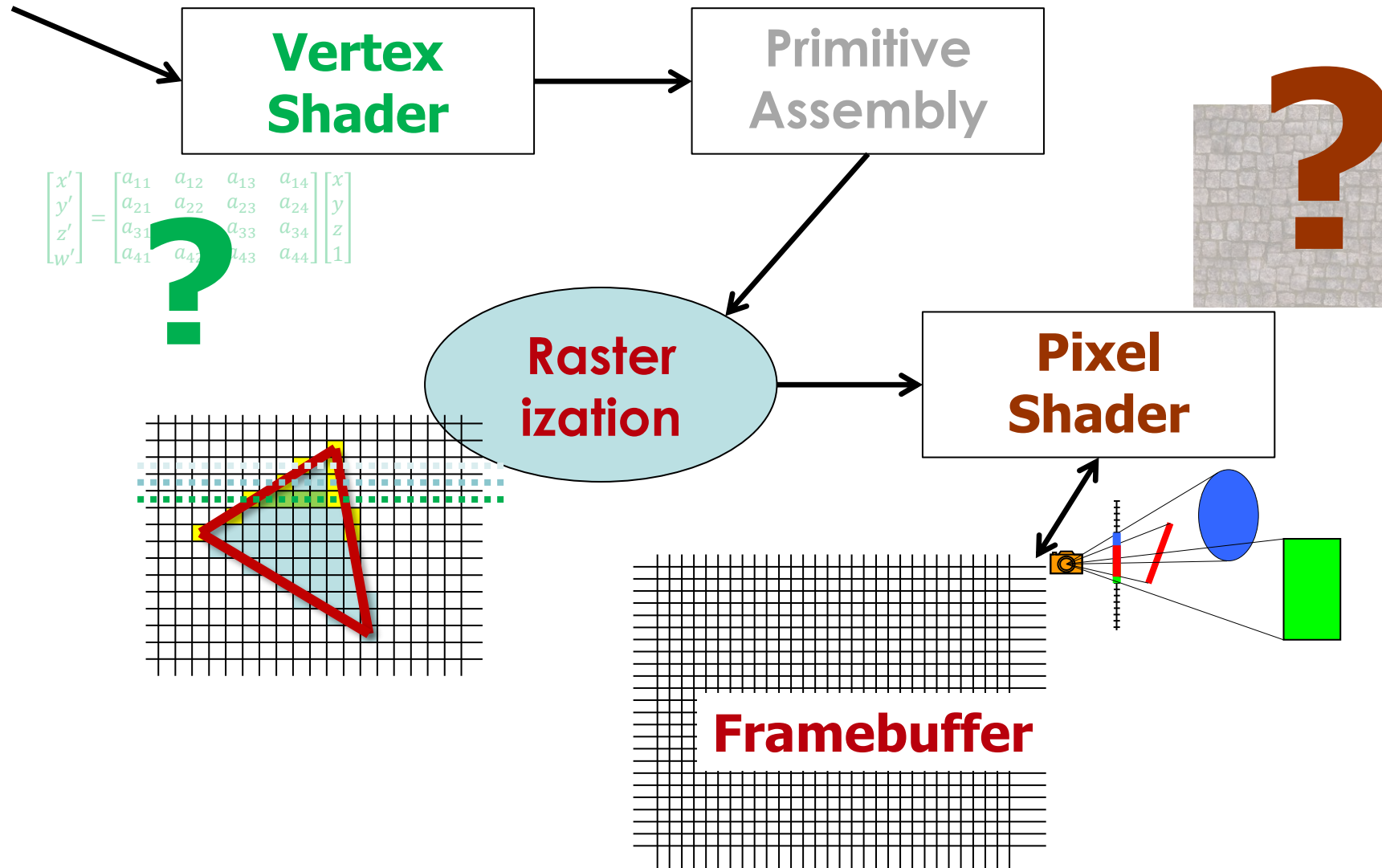
16bitů RGB

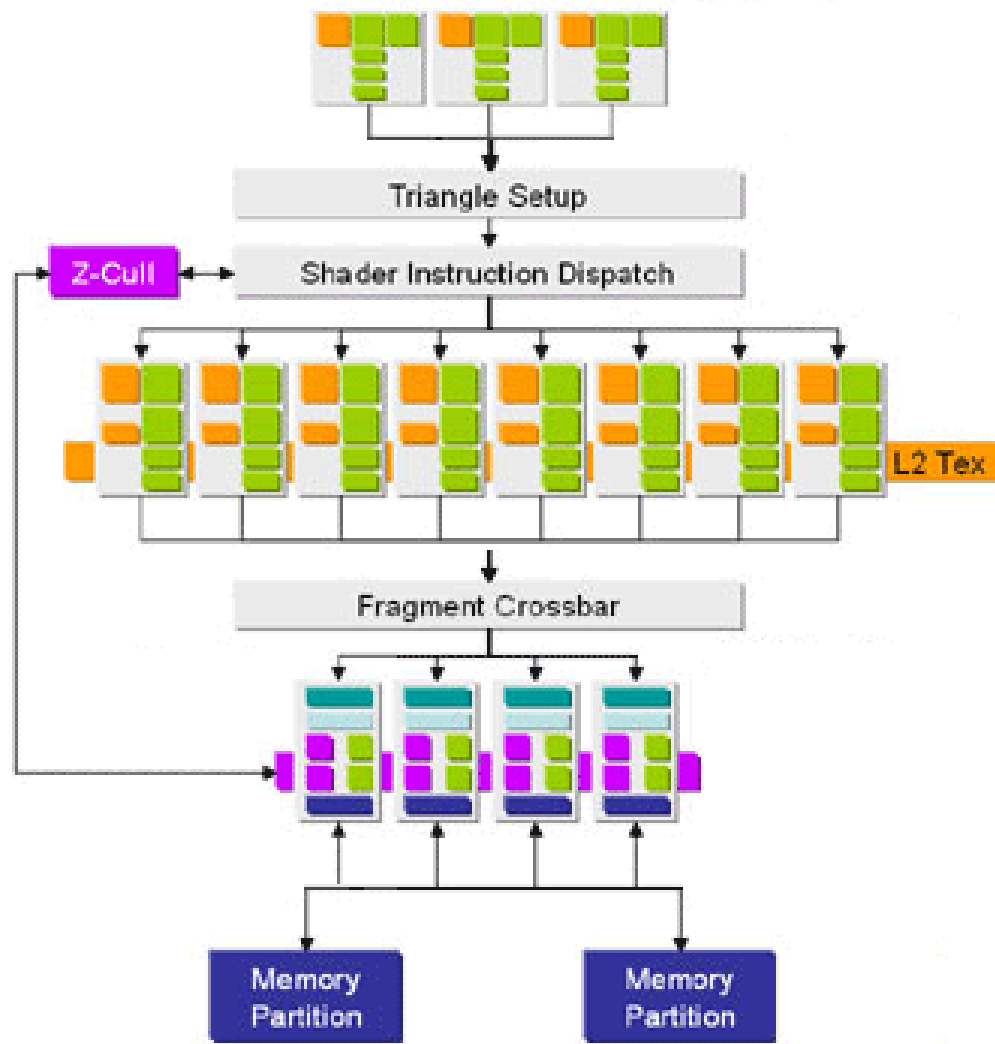


24+48 bitů - RGBD







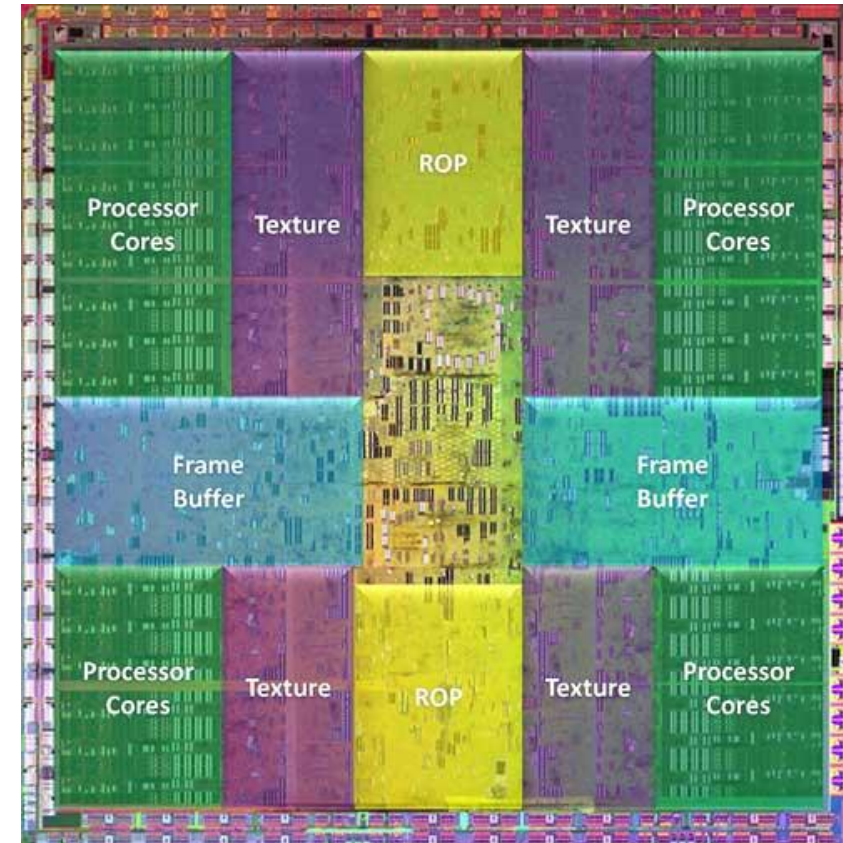
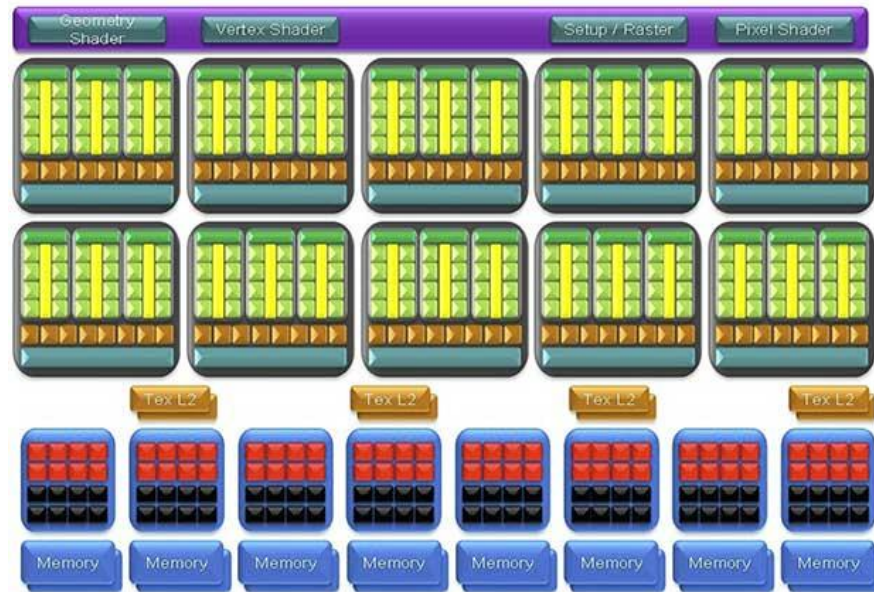




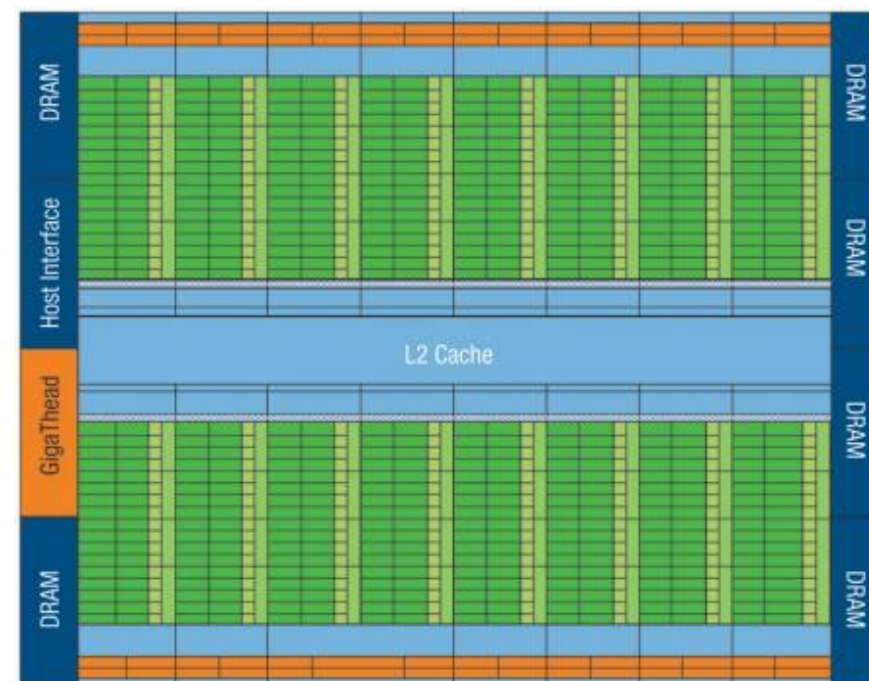
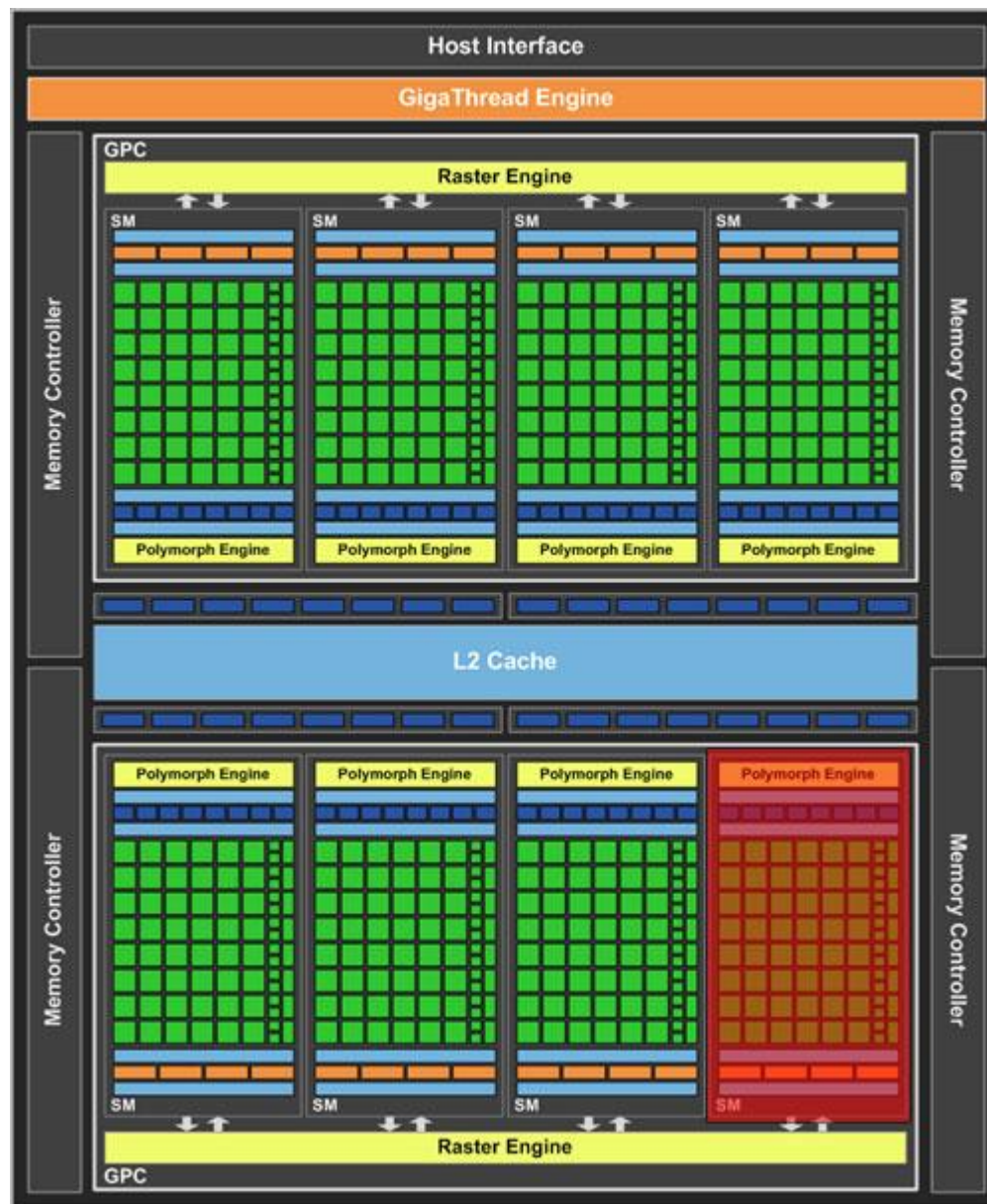
$$P = 1000 \times V$$

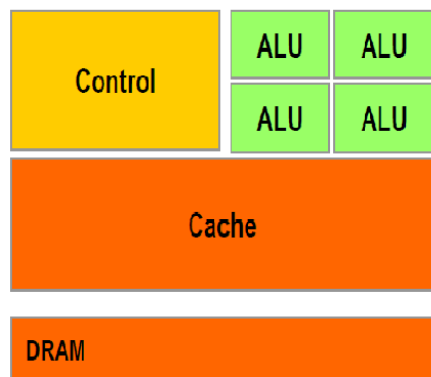
$$P = 5 \times V$$



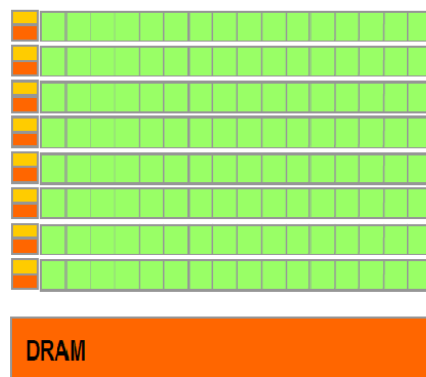


Struktura GPU (GTX 560Ti – cca 2010)

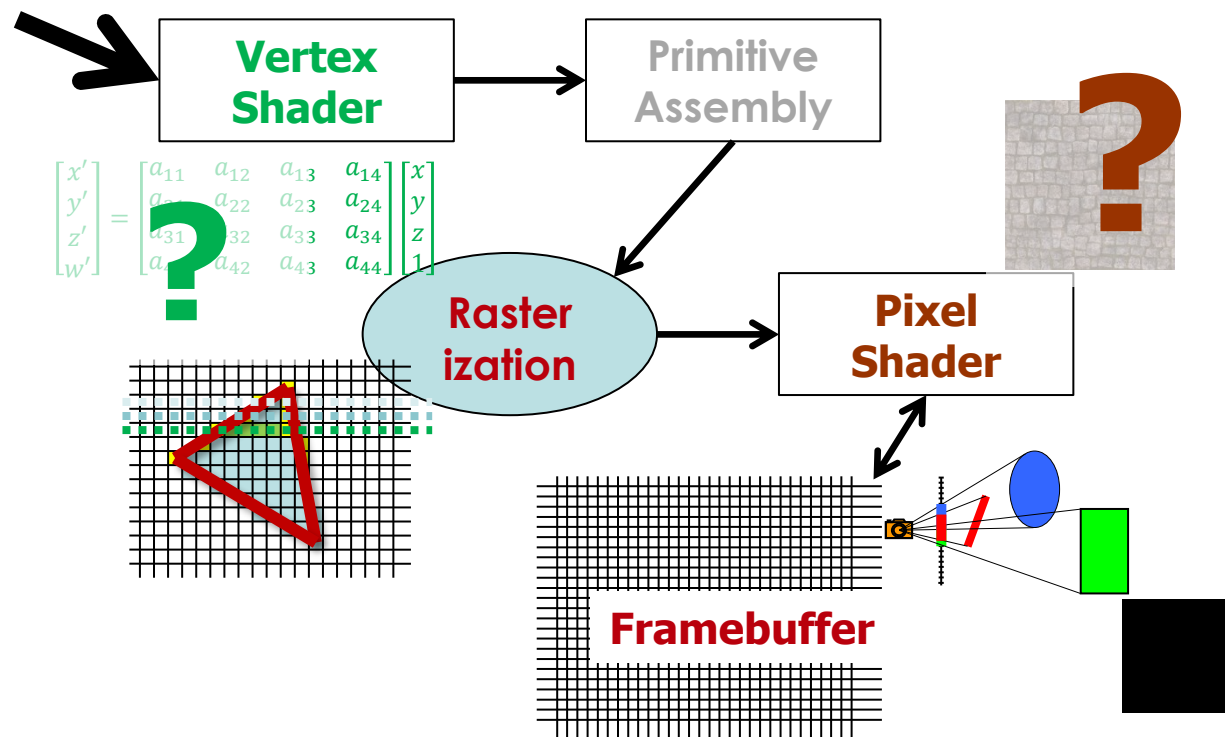




CPU



GPU



- Rasterizace
 - Úsečky
 - Trojúhelníka – interpolace!
- Graf scény
- Vykreslovací řetězec
 - Per-Vertex op. × Per-Fragment op. × Rasterizace
 - Framebuffer
 - Vertex Shader, Pixel Shader